

# HPS GPR study and plot logbook

Forty study and presentation entries; 26 selected plot pages with sources and speaking notes.

WHAT THE RECENT SERIES ESTABLISHED

Native signal MC motivates campaign-specific shape and window choices. Paired injections and clean-training controls reveal response loss that a lower observed upper limit can hide.

WHAT TO PRESENT FIRST

Use the 12-minute route on page 2. Pages 4-29 are reusable plot cards; the final pages locate the complete study packages and dated presentation snapshots.

WHAT REMAINS CONDITIONAL

Saved templates, null sources, kernel states and method choices define each experiment. These studies do not by themselves establish physical efficiency, unconditional coverage, a calibrated coupling exclusion or discovery.

## Independent local-to-global result

### 2016

B count: 89 / 1,024 | add-one p: **0.08780**

Exact 95% binomial interval: [0.07038, 0.10587]

### 2021 10%

B count: 231 / 1,024 | add-one p: **0.22634**

Exact 95% binomial interval: [0.20032, 0.25244]

### Common coupling

B count: 75 / 1,024 | add-one p: **0.07415**

Exact 95% binomial interval: [0.05804, 0.09095]

v6.4.4: fixed local maps from 1,024 A scans; independent 1,024 B scans. Each quoted interval conditions on A and the null source. Combined is a joint one-coupling fit. See P23-P26.

# A twelve-minute explanation

Eight stops, about 90 seconds each. Use the other cards to answer follow-up questions.

01 / P09 / PAGE 12

## Start with the selected signal-MC distributions.

Their broad tails and displaced cores motivate changes to the extraction model.

02 / P12 / PAGE 15

## Compare both shapes at a fixed expected yield.

Use independent pilot and evaluation cohorts; distinguish raw recovery from the paired increment.

03 / P17 / PAGE 20

## Show the controlled training-contamination experiment.

Removing only training-bin signal nearly restores direct-MC response in the displayed checks.

04 / P20 / PAGE 23

## Keep campaign-specific shape evidence.

2016 has a smaller core shift and must not inherit the 2021 law.

05 / P22 / PAGE 25

## Explain why a lower limit can be misleading.

Reduced response largely cancels the narrower window's smaller raw spread.

06 / P19 / PAGE 22

## Show what changes in the observed joint fit.

The common-coupling local minimum moves when the 2021 template changes; keep that release distinct.

07 / P23 / PAGE 26

## Give the independent whole-scan probability.

Combined  $p=0.07415$ , conditional on the fixed local maps, source and declared grid.

08 / P25 / PAGE 28

## Finish with the mass correlations.

Overlapping fits explain why raw grid counts and a single resolution count cannot replace coherent scan toys.

# Keep these distinctions visible

The versions are a chain of conditional studies, not repeated independent confirmations.

## RAW RECOVERY AND PAIRED RESPONSE

Raw recovery is  $\text{mean}(\hat{A}/A)$ . Paired response is  $\text{mean}[(\hat{A}_{\text{signal}} + \text{background} - \hat{A}_{\text{background}})/A]$ , using the same background toy. A null subtraction cancels an offset; it does not correct response.

## REFERENCE RESOLUTION AND MC CORE WIDTH

The original mask uses  $\pm 2.25 \sigma_{\text{ref}}$  around the hypothesis or shifted center. MC  $u = (\text{mass-core center})/\text{core width}$  is another coordinate. A value expressed in  $u$  cannot be treated as  $\sigma_{\text{ref}}$ .

## LOCAL AND GLOBAL ORDERING

v6.4.3 calibrates  $\max(\text{raw } q_0)$ . v6.4.4 freezes A local rank maps and calibrates  $\min(\text{local rank } p)$  with independent B. Their global probabilities answer different questions.

Finite ranks: zero exceedances gives an add-one floor and a binomial bound. An empty accepted yield grid is not a physical zero limit. Uncertainty bars retain the source study's definition; MC sampling uncertainty omits unmodeled systematic effects.

## EXPECTED YIELD AND REALIZED COUNTS

v6.2 fixes exact N and samples multinomial categories. v6.3.1 fixes expected  $A = z \cdot s_0$  from an independent pilot and lets Poisson counts fluctuate. Its pulls use expected A.

## FIXED SOURCE AND FITTED BACKGROUND

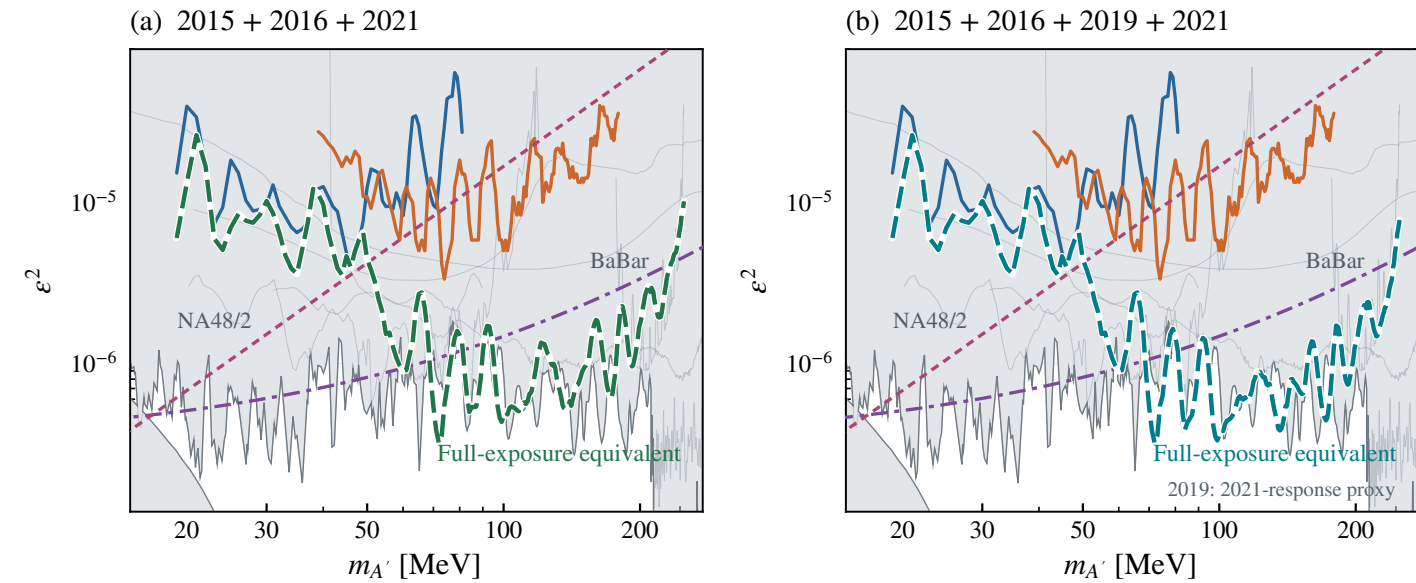
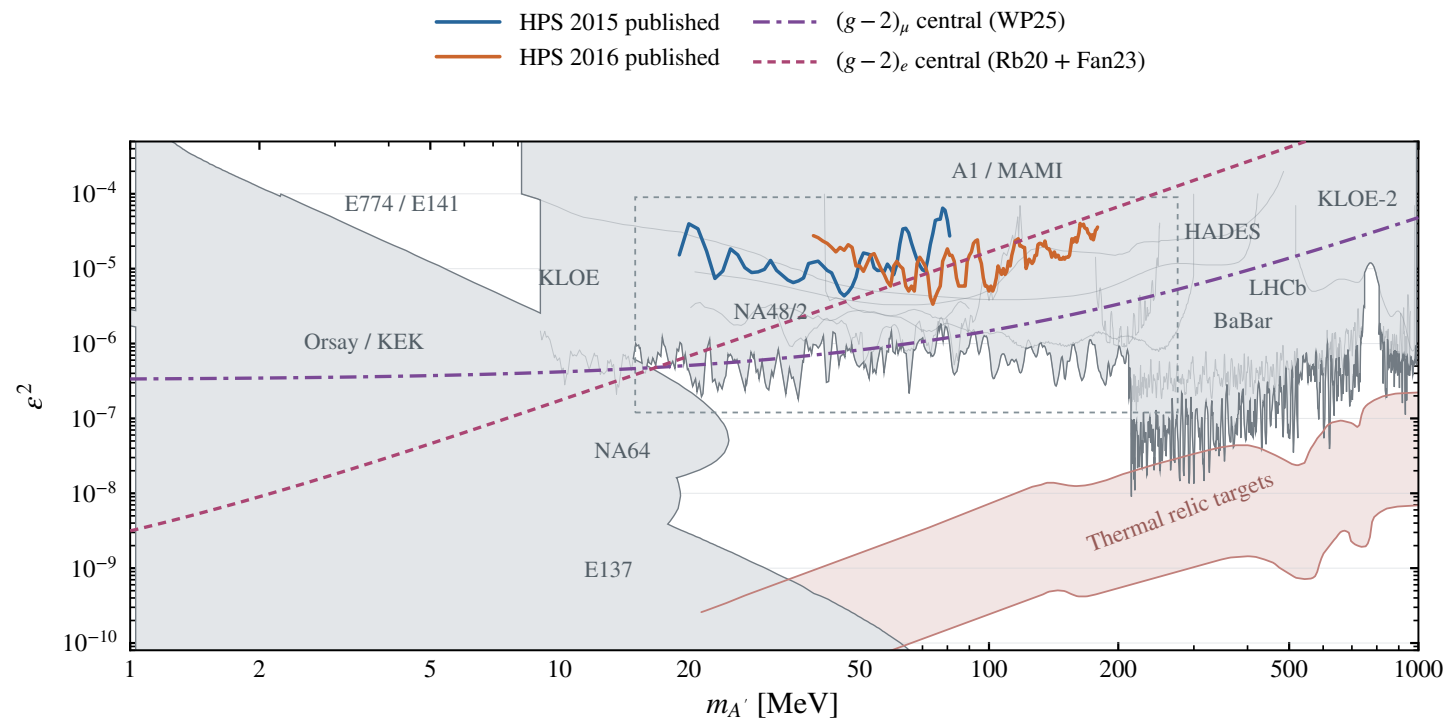
A fixed GP mean may generate the toys while each toy still recomputes its conditional GP prediction. Holding that prediction fixed inside the likelihood ( $C=0$ ) is a further, different change.

## COMMON COUPLING AND OTHER COMBINATIONS

The joint physical model uses one coupling with campaign-specific normalizations and nuisance blocks. Independent amplitudes, Fisher/Stouffer, and selecting among searches are separate hypotheses.

# Place the studies in the reach picture

The new comparisons start from the released samples and explicit exposure assumptions.



## READ THE PLOT

Compare the matched lower panels; the added-2019 curve uses a measured 1% high-psum spectrum with borrowed 2021 response/conversion inputs.

## SUGGESTED EXPLANATION

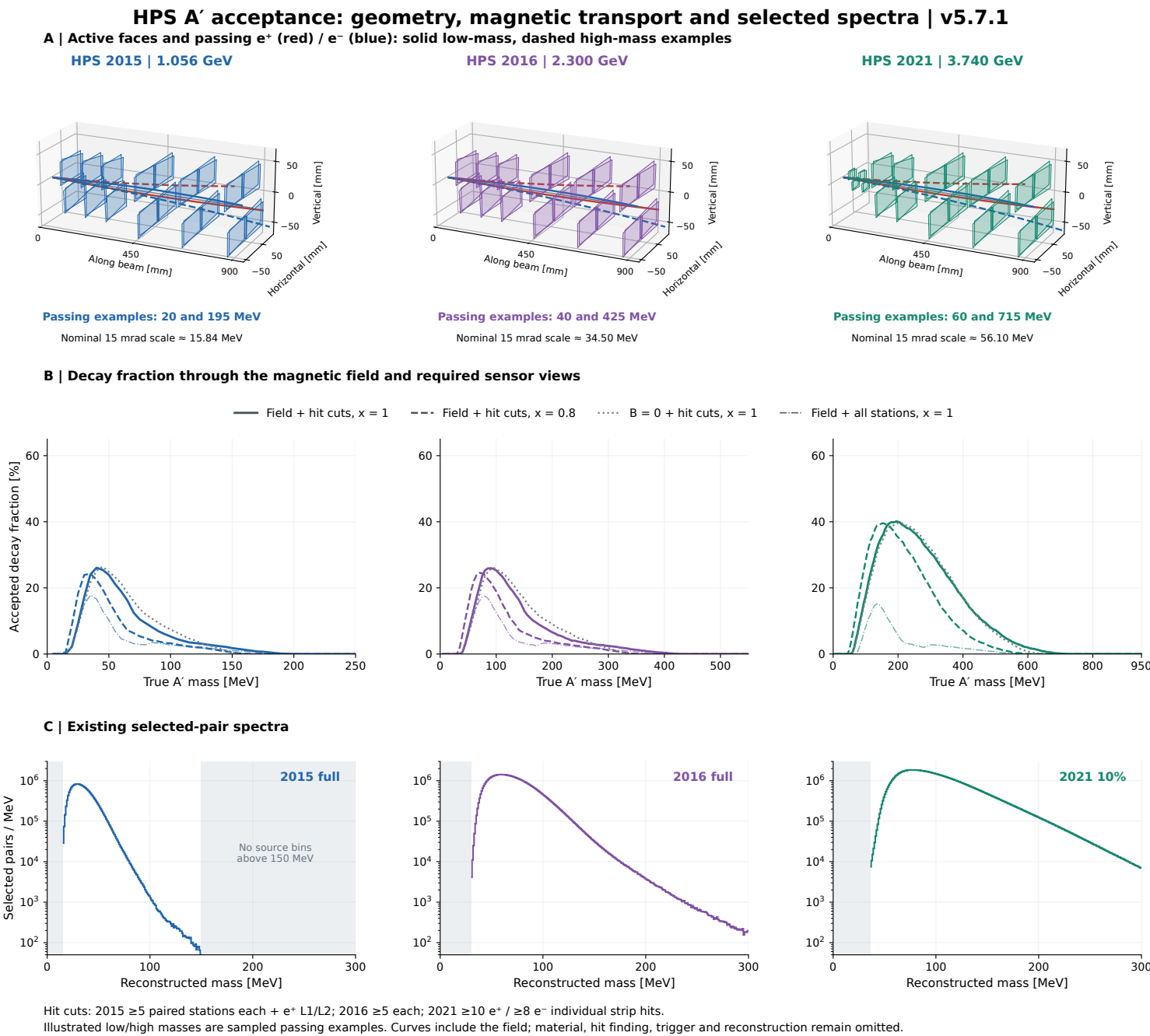
These curves show how the present samples map into an observed-equivalent full-exposure display.

## CLAIM BOUNDARY

Historical world-contour context and approximate confidence-level conventions are preserved. This is not a calibrated future expected-sensitivity band.

# Explain the beam-energy and geometry dependence

Magnetic transport and sensor-hit requirements replace the earlier outer-angle surrogate.



## READ THE PLOT

Read columns by campaign. Geometry, calculated acceptance and selected mass spectra answer different questions; their vertical scales are not interchangeable.

## SUGGESTED EXPLANATION

The detector geometry and boost help explain the accessible mass scales, while real efficiency requires more information.

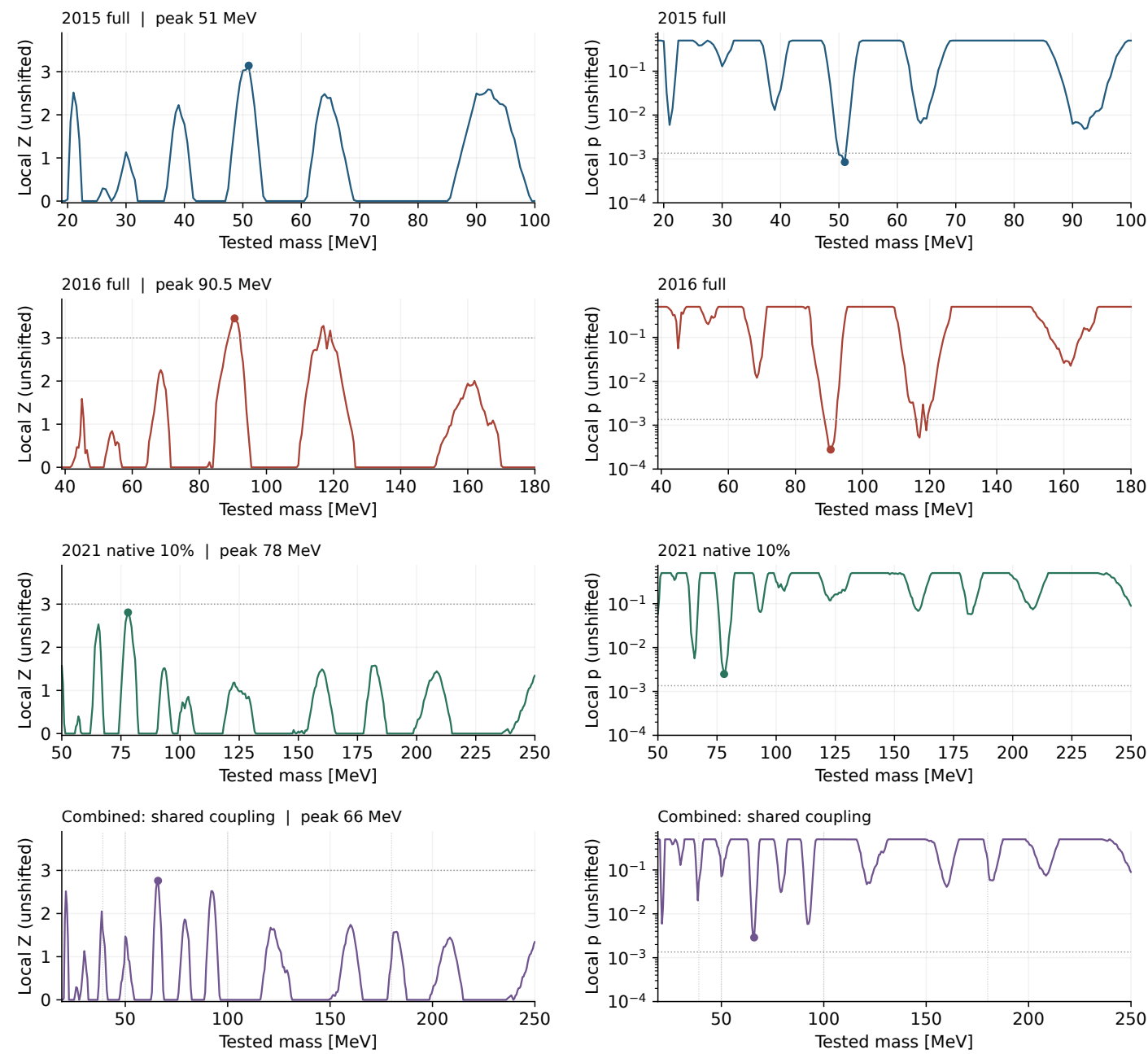
## CLAIM BOUNDARY

Conditional fixed-decay and hit fractions omit production weighting, reconstruction, trigger/selection efficiency and detector uncertainty.

# Keep the raw local excesses identifiable

The conventional local display uses the unshifted signed likelihood root.

Observed local curves: no reference centering or rescaling



## READ THE PLOT

$Z=\max(r,0)$  and  $p=\Phi(-Z)$ ; deficits display  $p=0.5$ . The 2016 peak is 90.5 MeV at  $Z=3.4525$ ; shared coupling peaks at 66 MeV at  $Z=2.7602$ .

## SUGGESTED EXPLANATION

These are local, asymptotic summaries of the observed fits. Their ranking is preserved before any reference remapping.

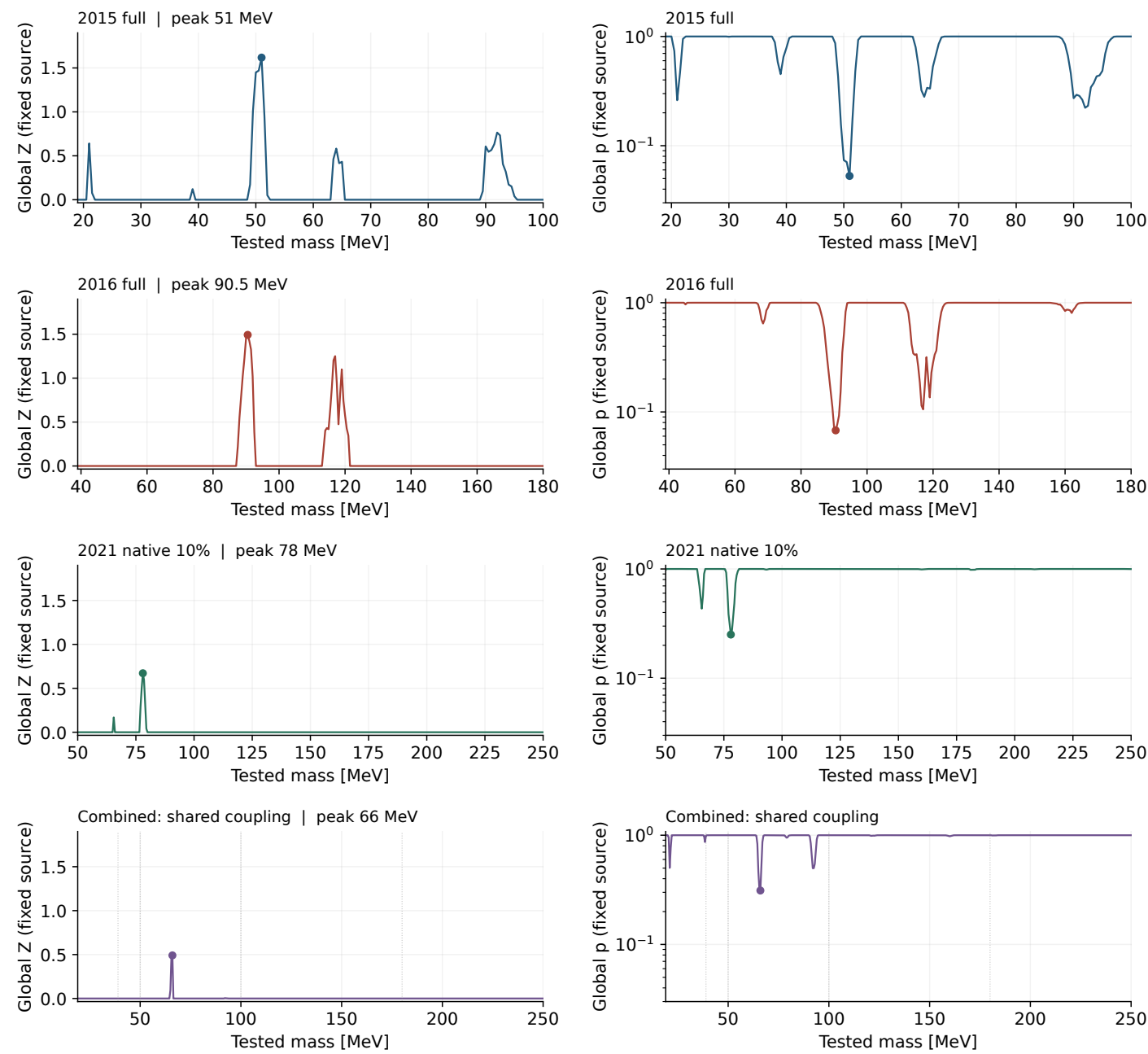
## CLAIM BOUNDARY

This is the September Gaussian baseline. Do not attach a global probability from a later MC window or different ordering.

# Show what the scan-wide correction changes

A local peak and the maximum over a mass scan have different null distributions.

Full-range tails of the raw maximum: conditional on the fixed source



### READ THE PLOT

The global reference uses the raw maximum and correlated whole-spectrum fluctuations. Follow each stated search domain and source definition.

### SUGGESTED EXPLANATION

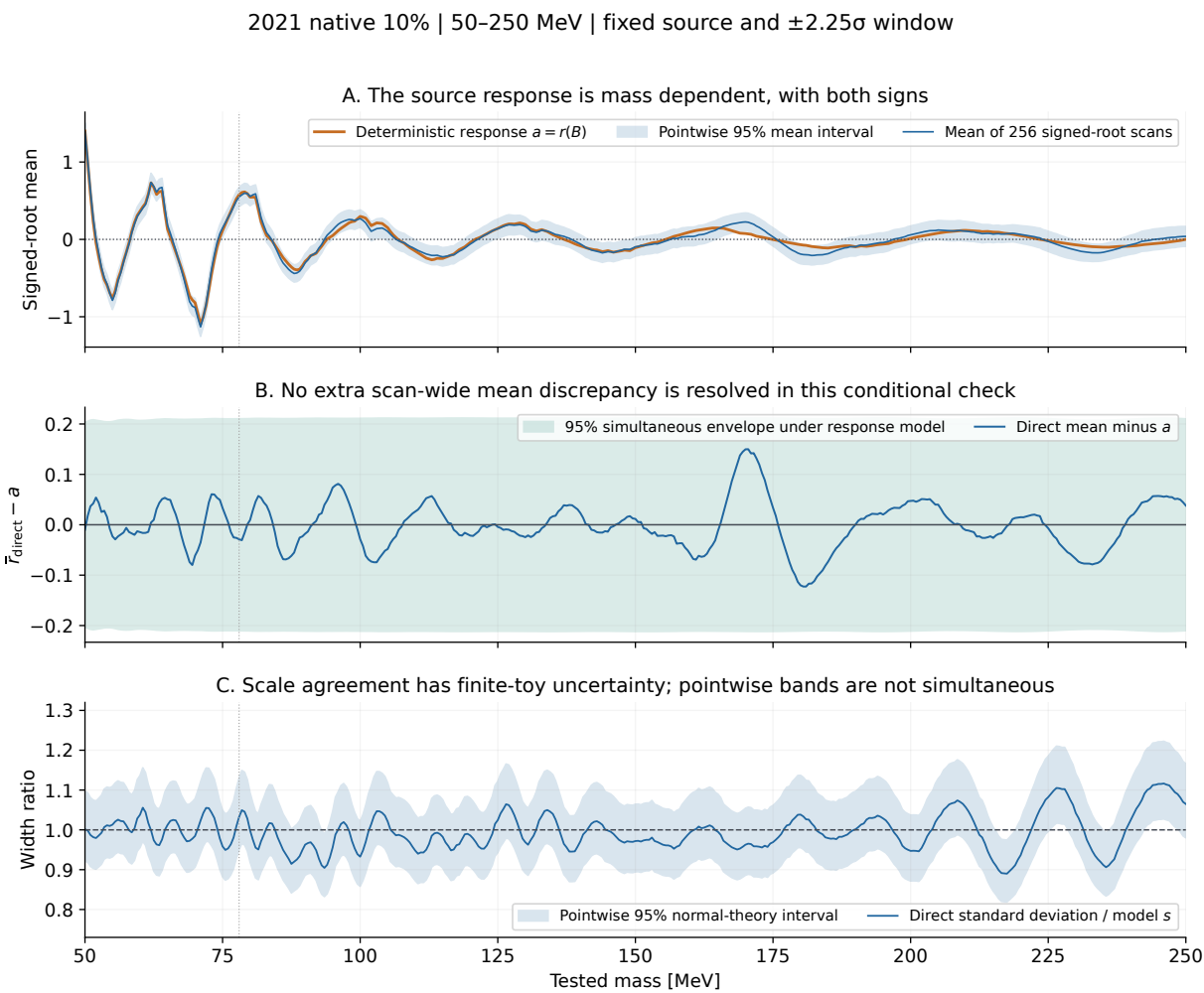
The probability must ask how often the whole specified search produces an excursion this large.

### CLAIM BOUNDARY

The global reference is conditional on fixed observed-derived background sources; prior method and source selection are not included.

# Separate a fixed-mass offset from a positive maximum

At 78 MeV the null signed-root mean is 0.548, a different effect from maximizing over mass.



### READ THE PLOT

Track the mass-dependent signed-root moments. A maximum tends to be positive even when individual signed estimators are centered; a nonzero fixed-mass mean needs its own explanation.

### SUGGESTED EXPLANATION

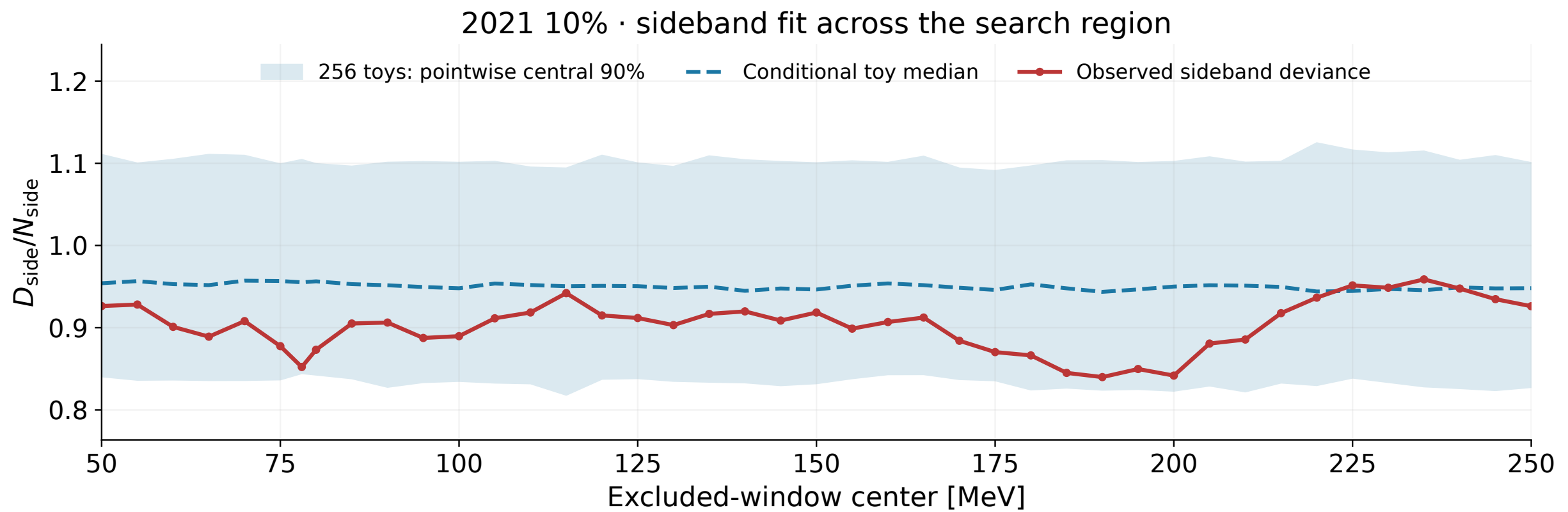
The selected peak cannot diagnose bias by itself. Here the fixed-source response also shows a local offset.

### CLAIM BOUNDARY

Moments and tails condition on the frozen generating spectrum; agreement with that source is not proof of physical background adequacy.

# Use the sideband check with its fitted-toy reference

All 42 observed values lie inside their pointwise central 90% conditional bands.



READ THE PLOT

Red is observed  $D_{\text{side}}/N_{\text{side}}$ ; dashed blue is the median from 256 repeated spectra and shading is the 5th-95th percentiles at each center.  $N_{\text{side}}$  counts scored bins, not fit degrees of freedom.

SUGGESTED EXPLANATION

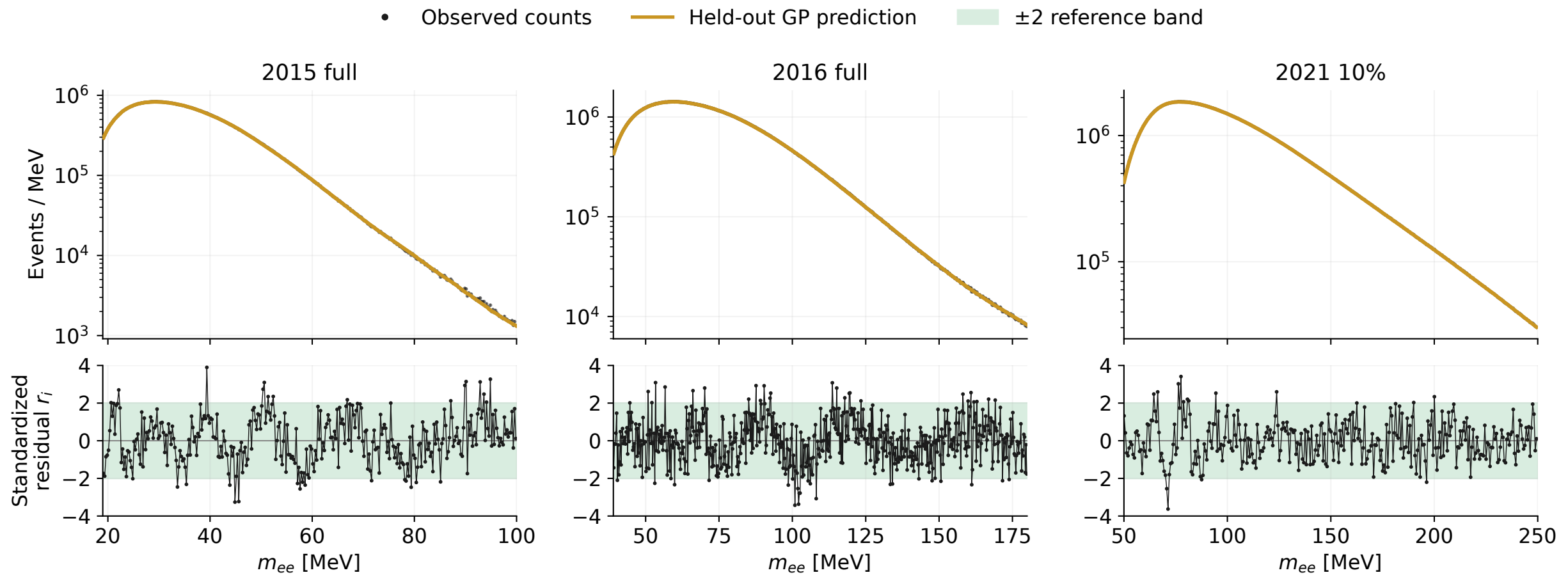
The reference median is about 0.95, so one is not an exact target after fitting.

CLAIM BOUNDARY

This scores fitted sidebands. Centers are correlated; pointwise agreement neither gives a global goodness-of-fit probability nor validates the excluded window.

# Explain the standardized residual panels

The residual is a local data-minus-GP difference scaled by counting and GP uncertainty.



READ THE PLOT

Each bin uses a prediction excluding its local signal window. Zero denotes agreement; positive values denote more data. The green +/-2 region is a visual reference.

SUGGESTED EXPLANATION

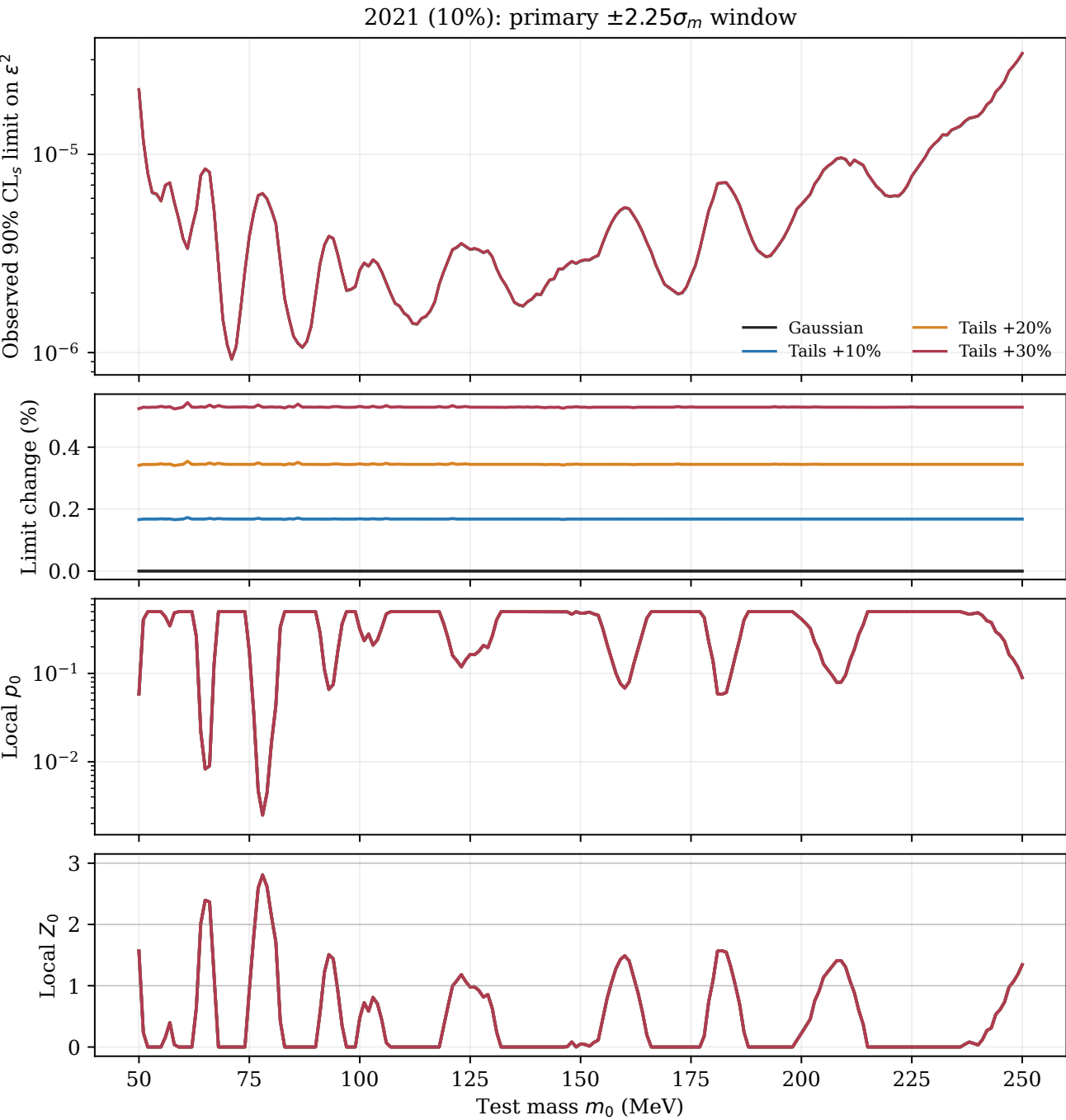
These panels make local behavior readable while retaining the fact that neighboring predictions overlap.

CLAIM BOUNDARY

The curves join local predictions, not one global fit. The green band is not calibrated simultaneous coverage.

# Keep a controlled tail test distinct from native MC

The controlled primary comparison changes tails while preserving the central unnormalized Gaussian shape.



### READ THE PLOT

Each alternative is compared to a freshly fitted Gaussian with the same  $\pm 2.25$  reference-resolution fit and exclusion mask. Full-support normalization is retained.

### SUGGESTED EXPLANATION

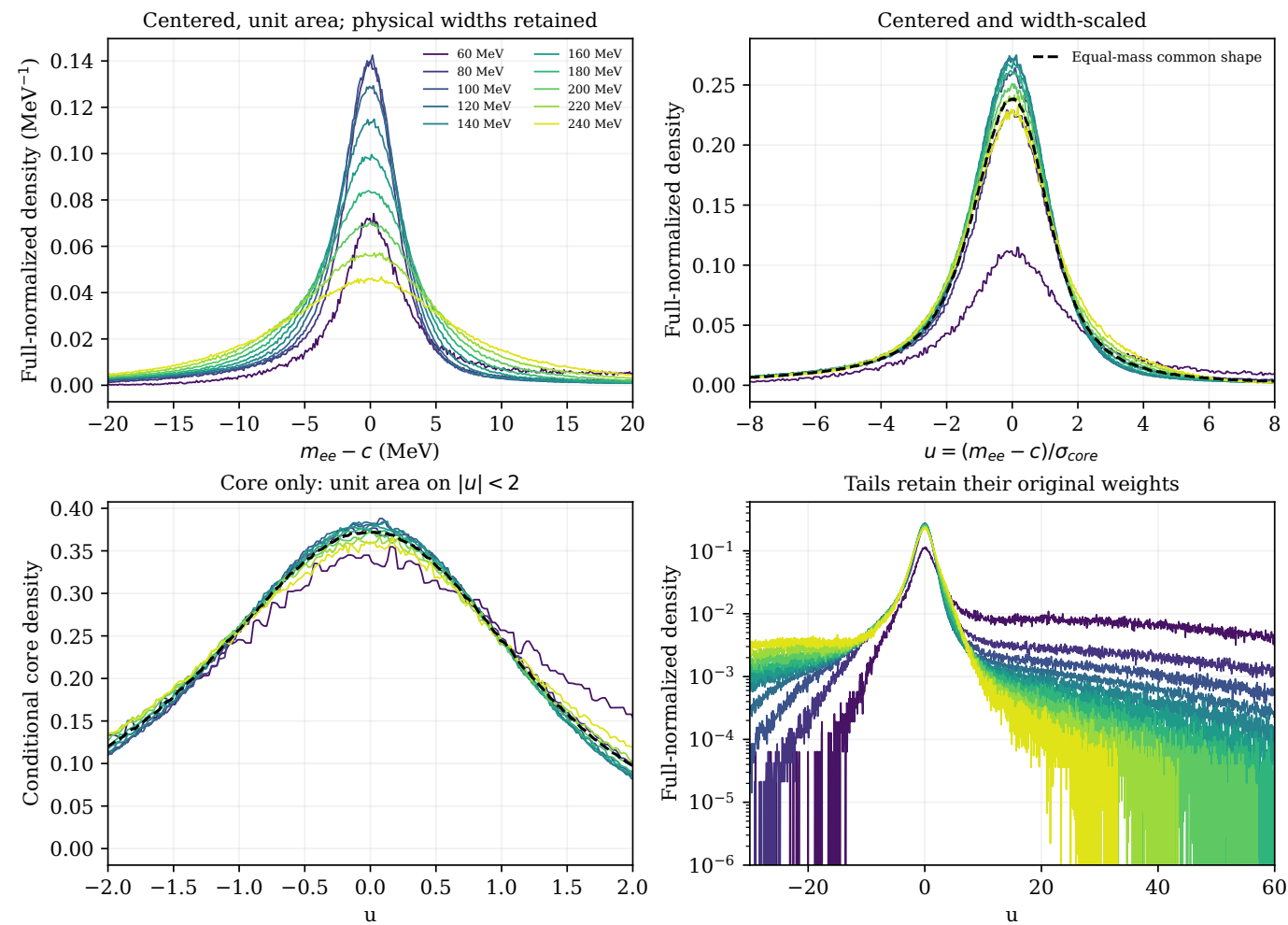
This isolates modest tail-family effects under a matched procedure.

### CLAIM BOUNDARY

It does not cover the broad displaced native MC shapes. Larger guards also change the fit bins and background prediction.

# Show what the native MC curves actually are

The colored curves are selected MC distributions, with fitted cores used to define coordinates.



## READ THE PLOT

Compare full distributions and the aligned core. A Gaussian center/width summary is not the empirical histogram; a common aligned shape is another model.

## SUGGESTED EXPLANATION

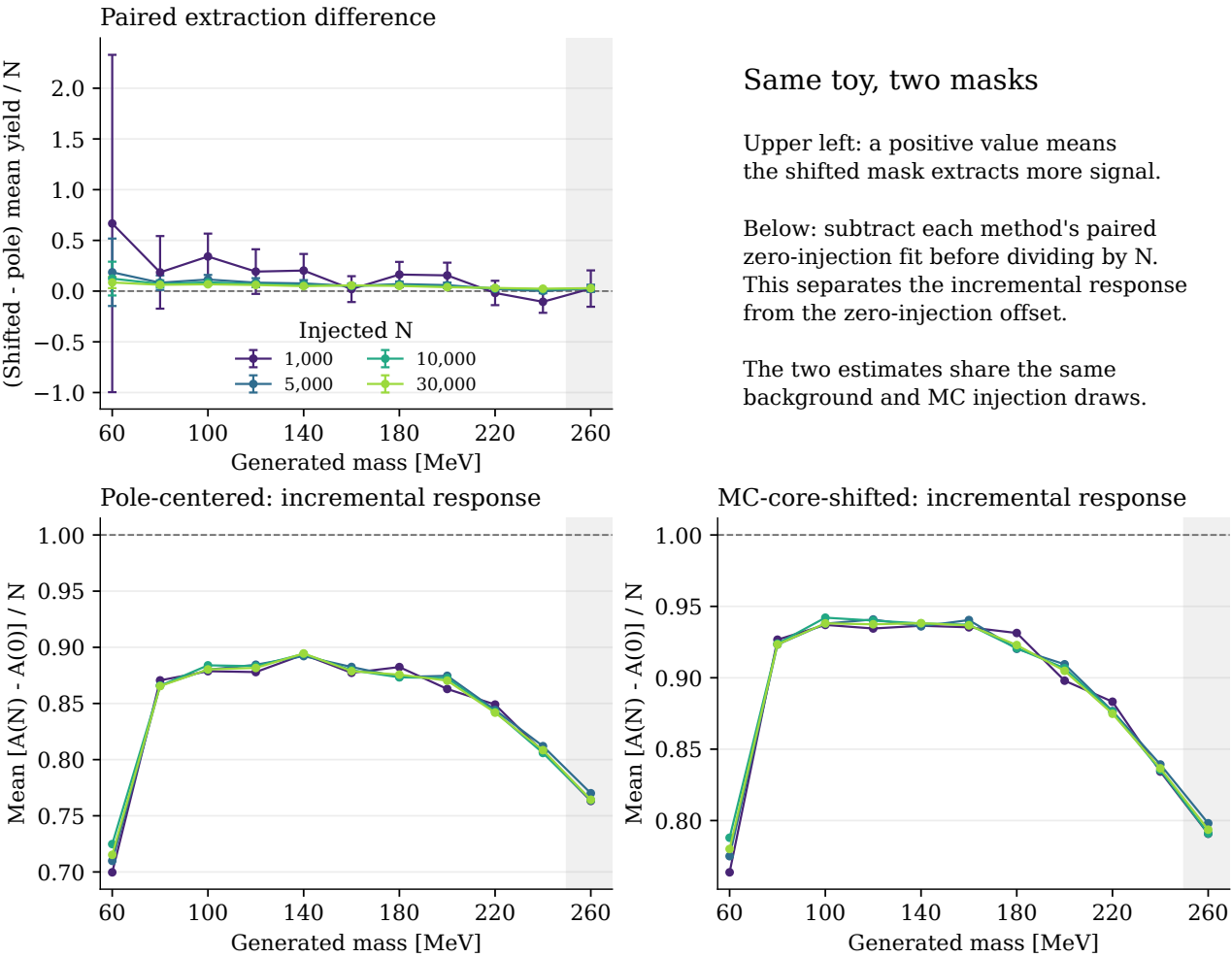
The MC contains structure outside a narrow Gaussian core, so normalization and the training region matter.

## CLAIM BOUNDARY

Selection equivalence and pure resonance association are not established for this supplied catalogue; the common shape remains conditional.

# Measure the signal-induced change with paired toys

Final 40-toy results show better response after shifting the window to the MC core.



READ THE PLOT

At N=30,000, paired response spans 71.5-89.5% for pole windows and 78.0-93.8% for shifted windows. Pairing subtracts the same toy's zero-signal fit; it does not redefine the primary fitted yield.

SUGGESTED EXPLANATION

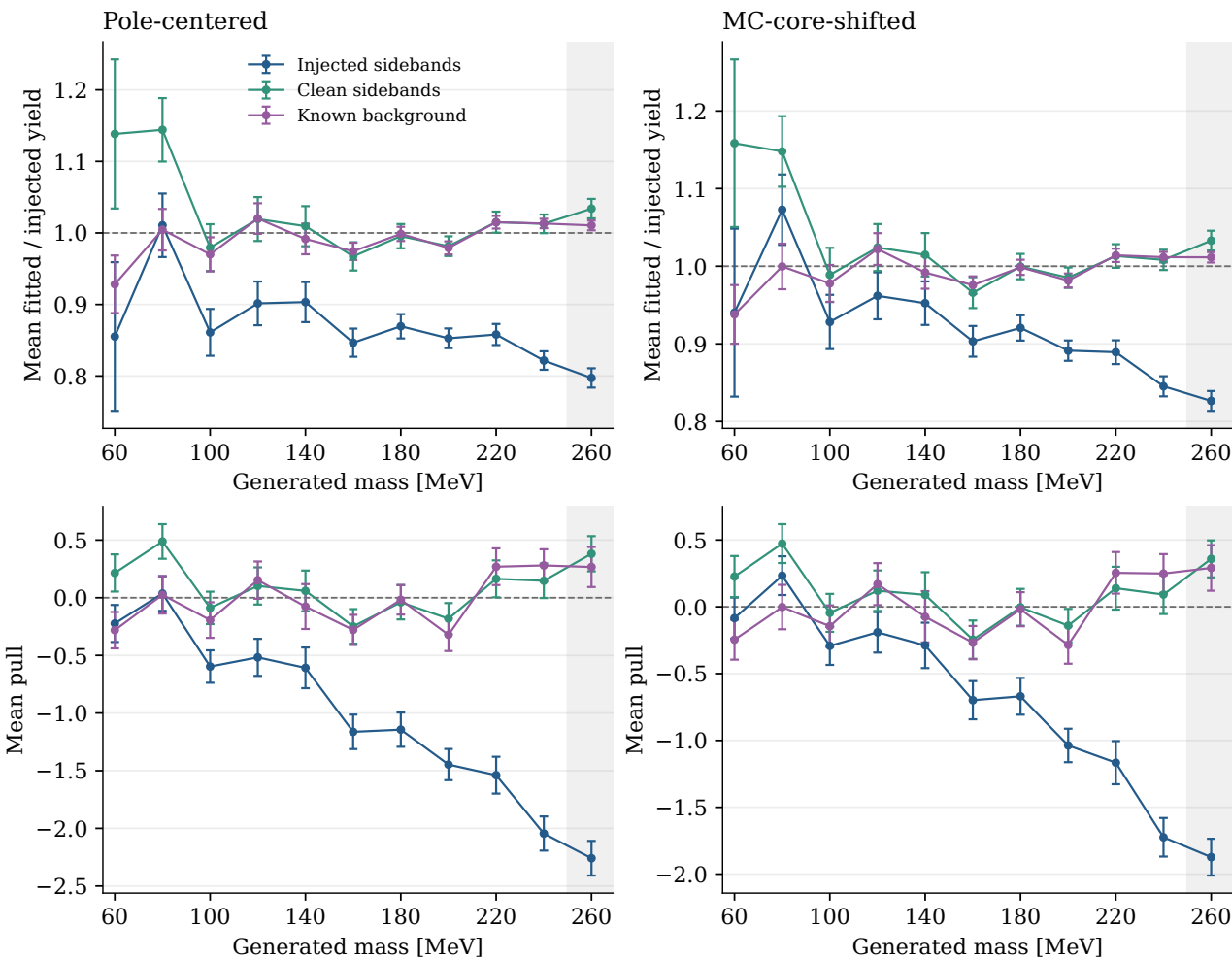
Centering helps, but it does not recover every injected selected candidate.

CLAIM BOUNDARY

These are exact-N injections with full-category multinomial draws. The 260 MeV extension uses the archived 250 MeV kernel anchor.

# Use controls to locate the response loss

Signal in GP training bins can be absorbed into the predicted background.



READ THE PLOT

Compare the contaminated-sideband result with clean-sideband and known-background controls at the same injected yield. Distinguish raw fitted/injected yield from the increment over the paired null fit.

SUGGESTED EXPLANATION

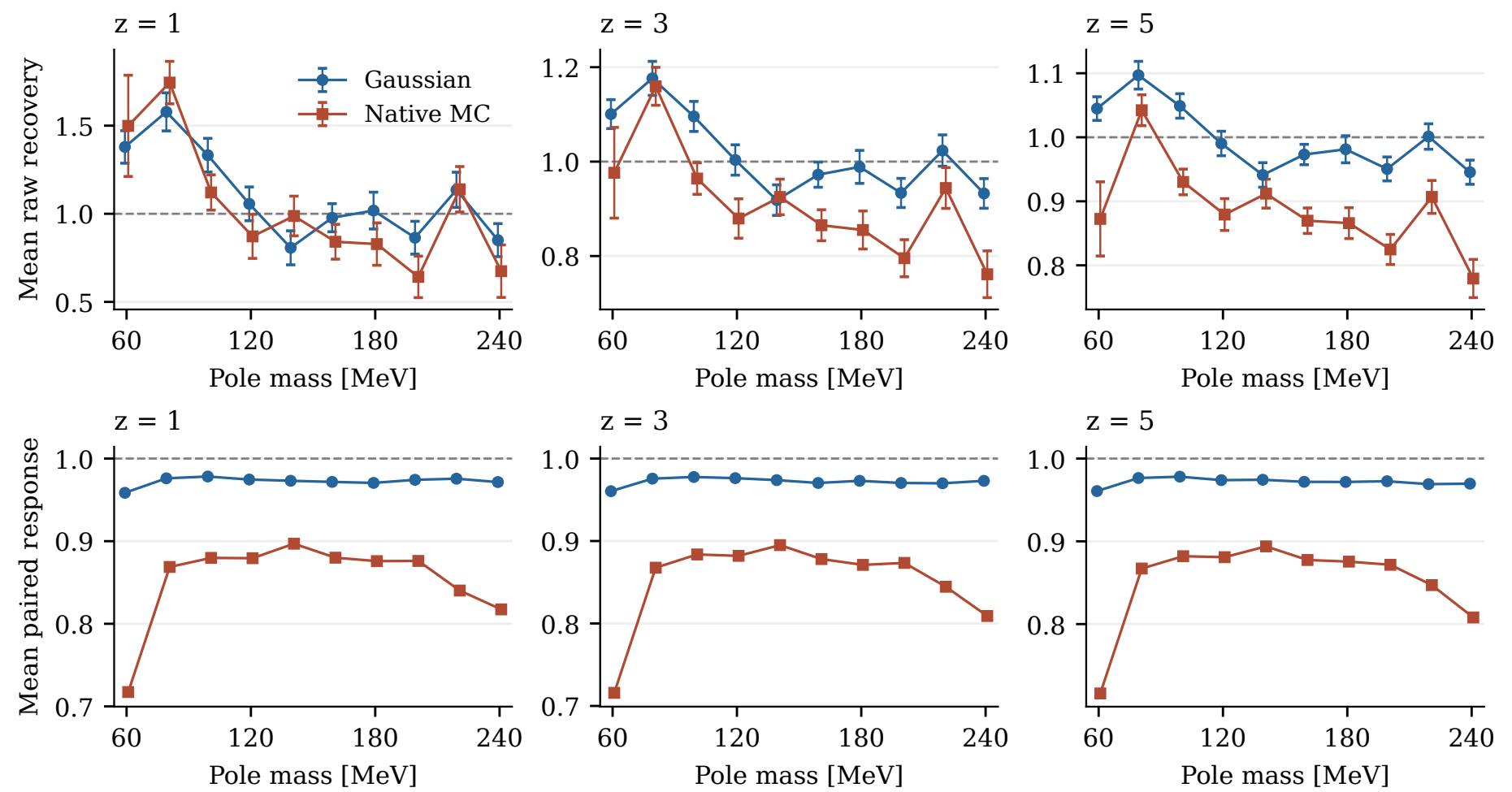
A fit can converge and still recover too little signal; the controls identify why.

CLAIM BOUNDARY

Clean training and a known background are diagnostic oracles, unavailable for a real unknown signal. Forty-toy containment is not coverage certification.

# Compare shapes at a common expected signal count

At z=5, paired response is 0.961-0.978 for Gaussian and 0.716-0.894 for native MC.



### READ THE PLOT

An independent 100-background Gaussian pilot fixes  $s_0(m)$ ;  $A=z*s_0$  is shared by shapes. The 100 evaluation backgrounds are paired, with independently fluctuating Poisson signal totals.

### SUGGESTED EXPLANATION

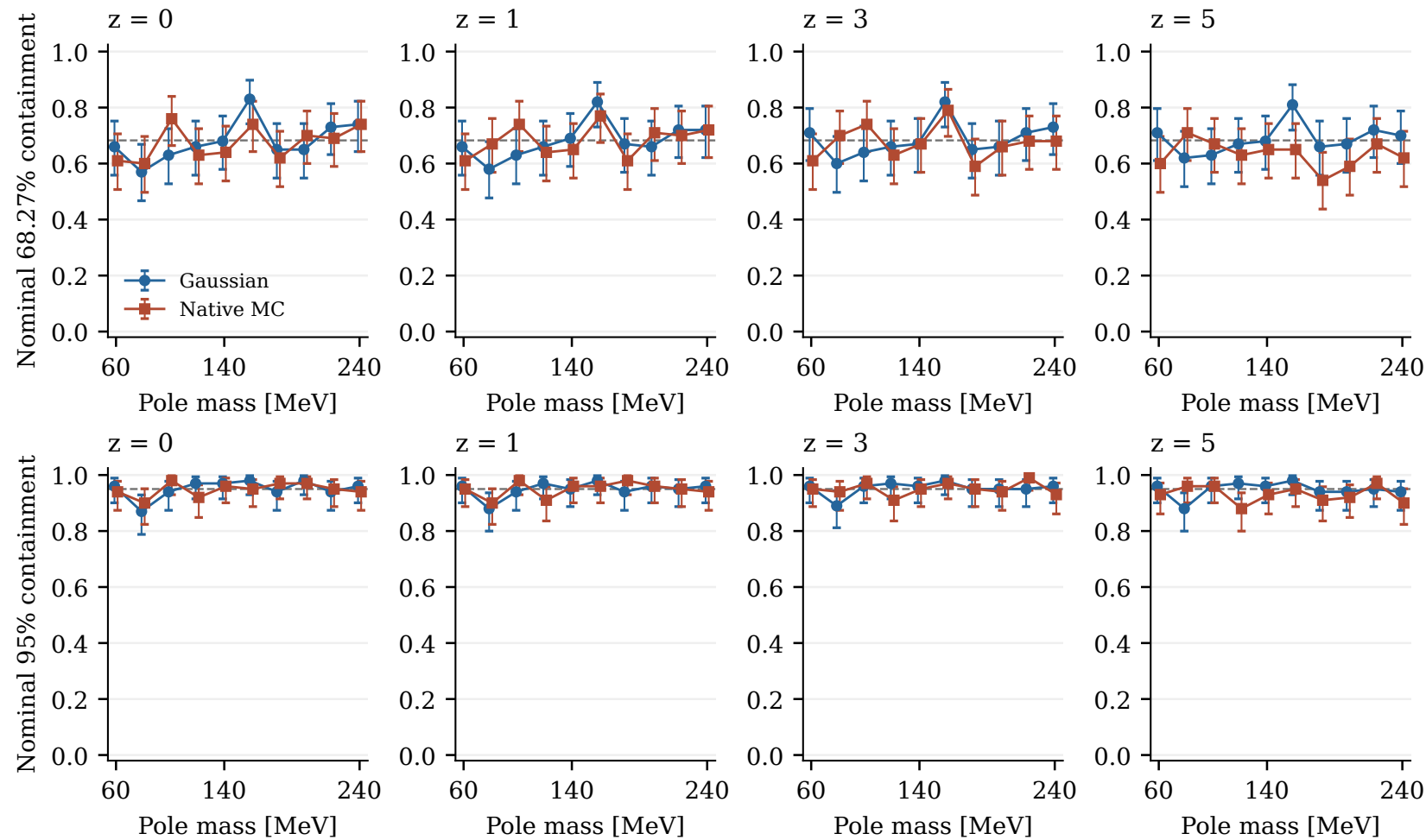
Equal expected signal counts do not imply equal extraction difficulty.

### CLAIM BOUNDARY

The pull truth is expected A, not the realized count. Curves summarize correlated cells, not independent measurements to average together.

# Report containment with its finite denominator

The z=5 nominal 95% signed sets contain truth in 88-98/100 Gaussian and 88-97/100 MC trials.



### READ THE PLOT

These sets use the truth-fixed profile statistic and retain every attempted toy. Exact two-sided 95% Clopper-Pearson intervals quantify count uncertainty.

### SUGGESTED EXPLANATION

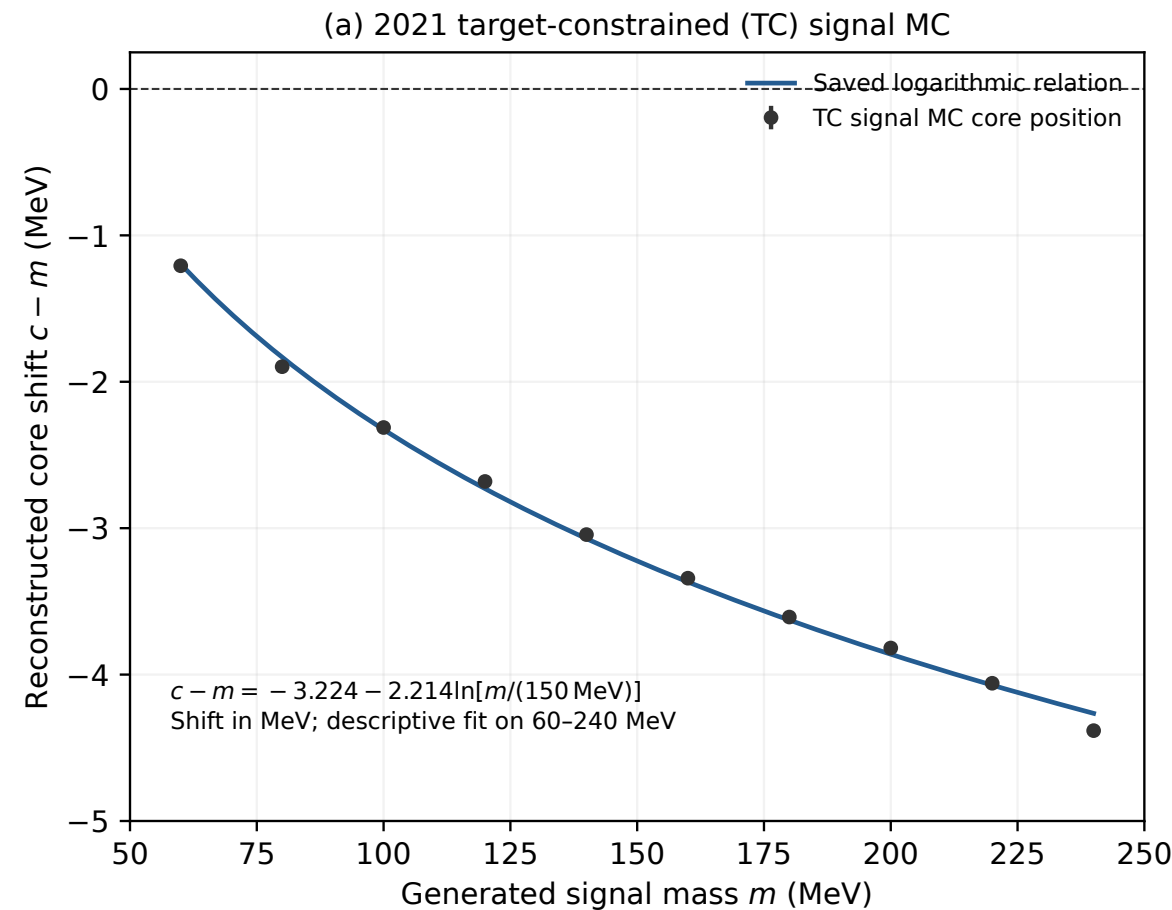
Completeness of the run and scientific closure are different: valid fits can still show non-nominal containment.

### CLAIM BOUNDARY

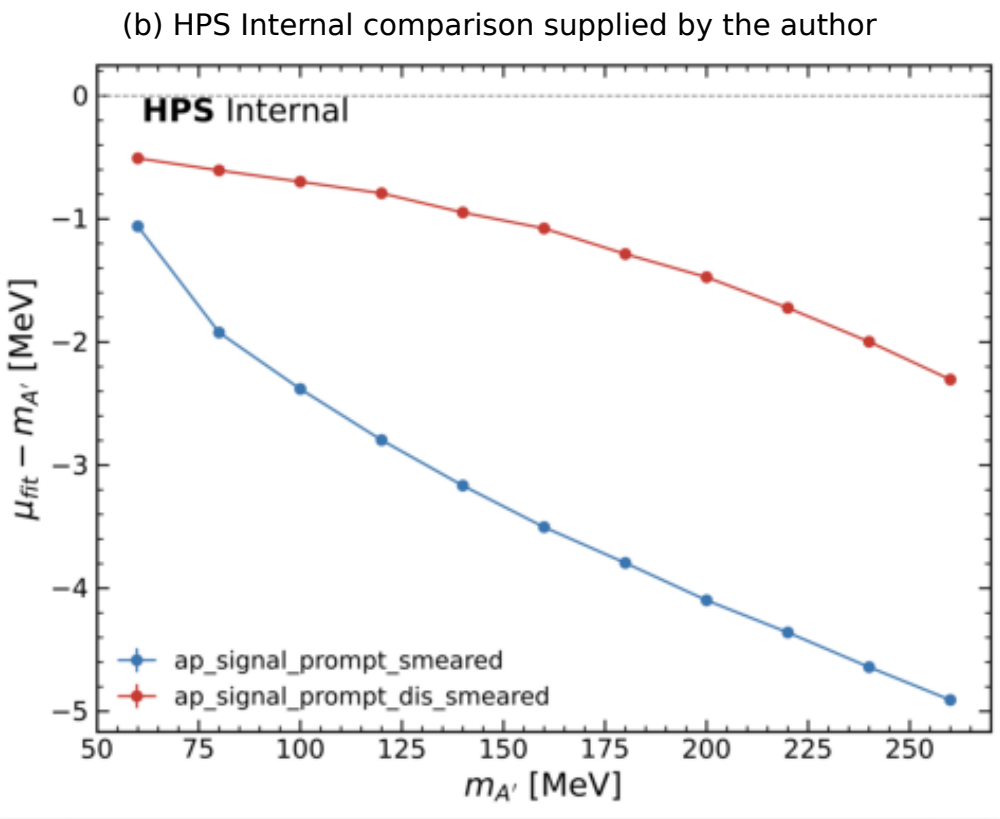
Signed profile-set containment is not a calibrated physical nonnegative-signal upper limit or unconditional coverage.

# Motivate the MC core-center correction

The reconstructed core shift should be measured from signal MC before observing fit performance.



Left bars: core-position SD from 32 signal-MC bin-resampling replicas (often smaller than markers).  
Right panel: original HPS image; its error definition was not supplied. Qualitative comparison only.



## READ THE PLOT

The measured core shifts are compared with the stated logarithmic law and the retained user-provided HPS image. MC bars are 32-replica bin-resampling standard deviations.

## SUGGESTED EXPLANATION

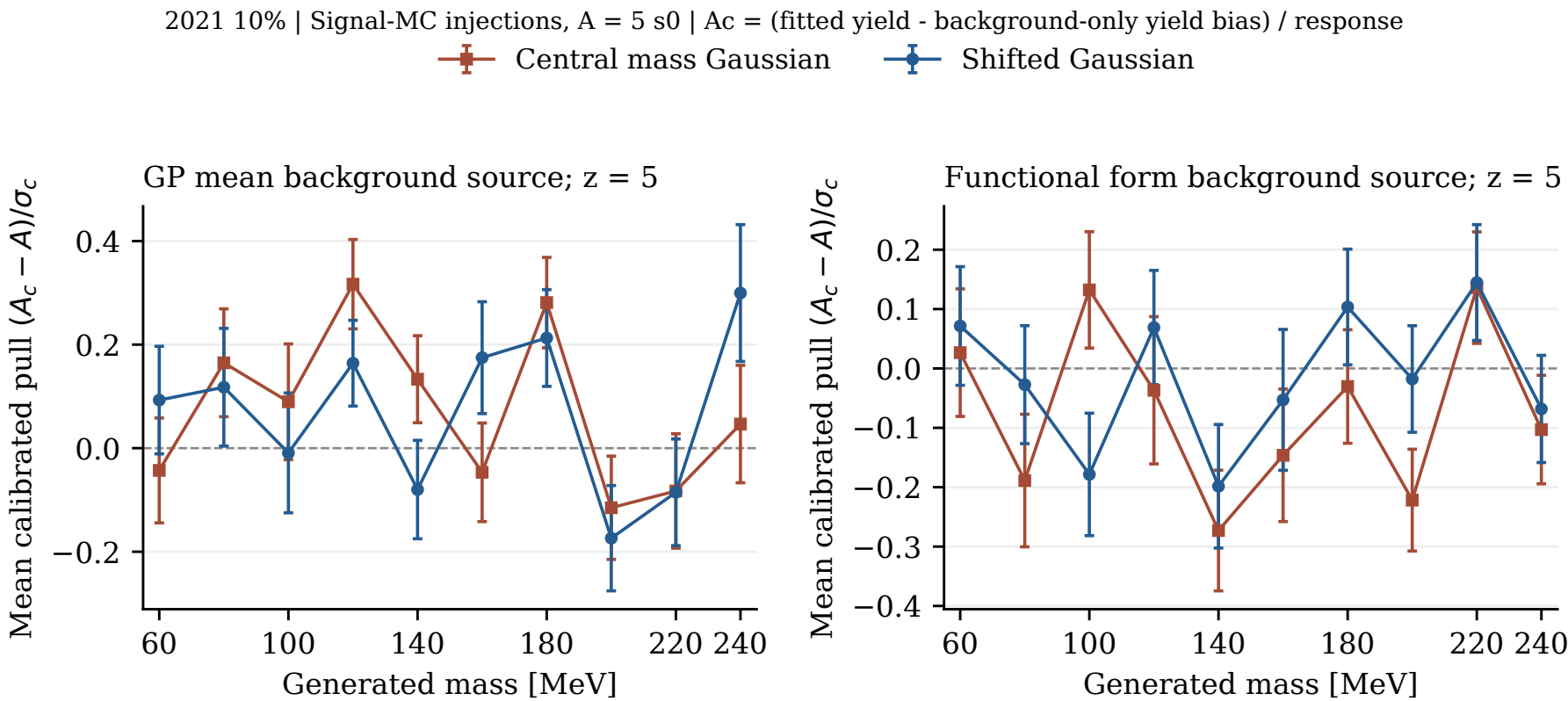
A reproducible core center gives a well-defined window shift, but does not fix the tails or signal response.

## CLAIM BOUNDARY

The retained reference image's error definition was not supplied. The 260 MeV catalogue point is shape-only and outside the logarithmic-law domain.

# Separate offset correction from response calibration

A frozen affine correction addresses the null offset and signal response together.



Means: sample standard error (sample SD / sqrt(100)). Widths: bootstrap standard error.  
Bootstrap: 2,000 whole-toy resamples keep masses, strengths and Gaussian choices together; sources use separate streams.  
Binomial fractions: all 100 trials retained; exact two-sided 95% Clopper-Pearson intervals.  
Mean bars after calibration condition on the frozen calibration table; its uncertainty is assessed separately.

### READ THE PLOT

$A_c = (\hat{A} - \Delta)/R$ . The panels show mean calibrated pulls  $(A_c - A)/\sigma_c$  at  $z=5$  for two separately calibrated background sources. Bars are standard errors across 100 independent evaluation toys; calibration-table uncertainty is assessed separately.

### SUGGESTED EXPLANATION

Subtracting a null mean cannot restore response. A correction needs both ingredients and an independent evaluation.

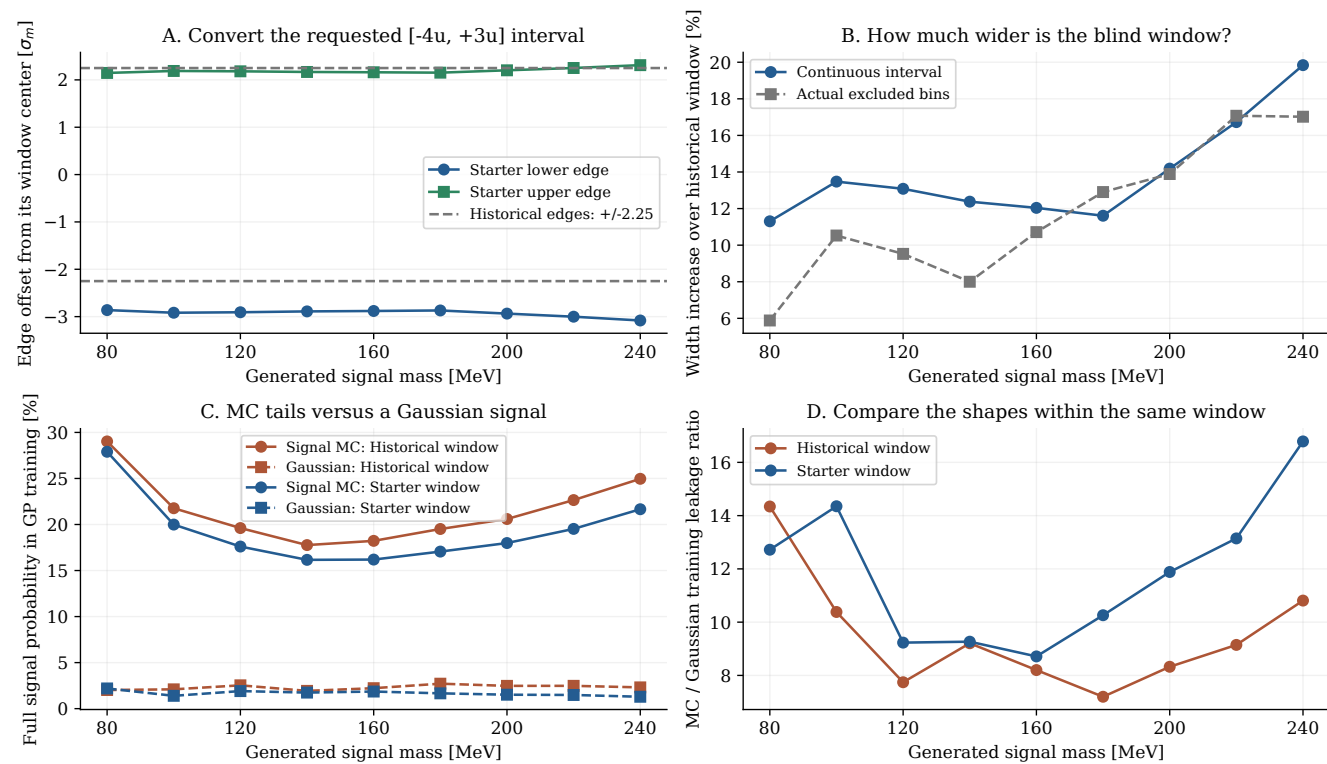
### CLAIM BOUNDARY

These are source-specific diagnostic corrections, not replacements for the primary finite-grid limits or evidence of transfer between background sources.

# Quantify the signal still entering GP training

The starter window is 11.3-19.8% wider, yet 16.16-27.90% of full selected MC remains in training.

2021 v16 TC signal MC: historical  $\pm 2.25\sigma_m$  versus requested  $u \in [-4, 3]$



How to read:  $u = (\text{reconstructed mass} - \text{core center}) / \text{core width}$ ;  $\sigma_m$  is the historical Gaussian analysis width.  
Interior core parameters are predicted from neighboring samples with the tested mass omitted, matching v6.3.7; endpoints use their measured cores.  
Gaussian curves use the shifted Gaussian with  $\sigma_m$ . Leakage counts only GP training bins within the recorded background spectrum.  
All probabilities retain the full selected signal normalization. Probability outside the spectrum is saved separately. No uncertainty bars: fixed-template geometry.

## READ THE PLOT

Compare the old  $\pm 2.25$  reference-sigma mask with  $[-4, +3]$  fitted-core-width units over 80-240 MeV. The MC training fraction drops only 1.12-3.30 percentage points.

## SUGGESTED EXPLANATION

A wider exclusion reduces leakage, but a Gaussian estimate substantially understates these MC tails.

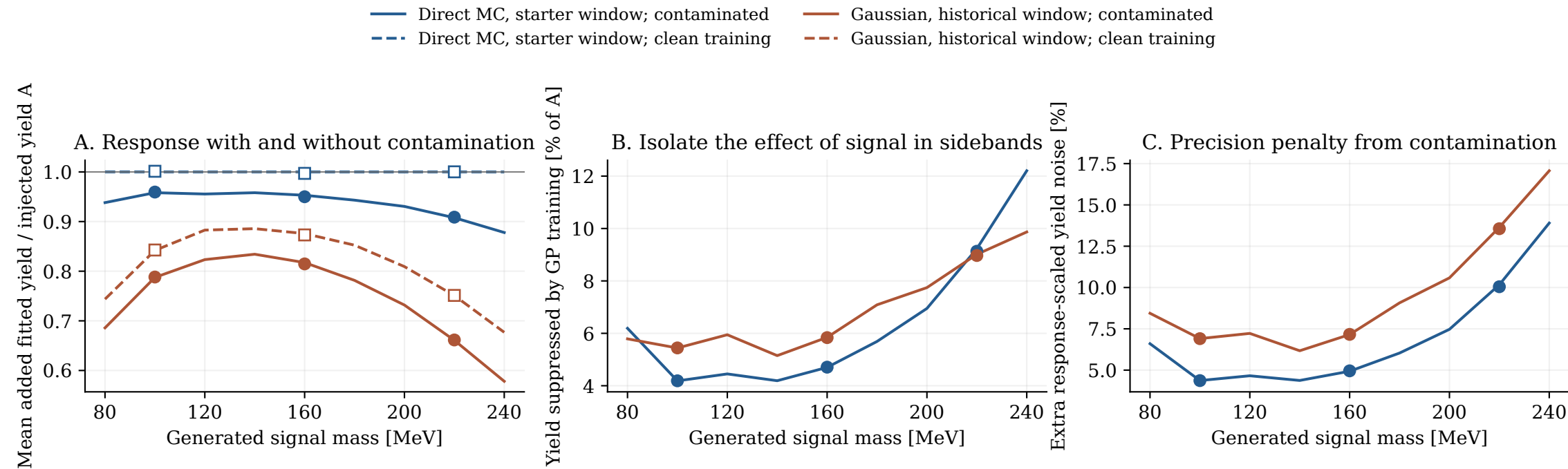
## CLAIM BOUNDARY

Fitted core width and reference resolution are different. Leakage probability is not identical to lost fitted yield.

# Demonstrate the effect of training contamination

At 100/160/220 MeV, clean training changes direct-MC response from 0.959/0.950/0.909 to 1.001/0.997/1.000.

2021 10% | GP mean background source | Full v16 TC signal-MC injection, A = 3 s0



How to read: clean training removes only the injected signal from GP training bins; the signal-fit counts, template and window are identical. Lines: deterministic mean-count spectra at nine masses, not toy means. Markers: 100 paired saved evaluation toys at 100, 160 and 220 MeV. A/B bars: sample standard errors of paired means. C bars: 95% intervals from 2,000 whole-toy bootstrap resamples; masses/methods stay paired. Panel B =  $\text{mean}(\text{Ahat\_clean} - \text{Ahat\_contaminated}) / A$ . Panel C =  $100 \times (\text{R\_clean} / \text{R\_contaminated} - 1)$ ; background-only noise is unchanged. s0 is the archived Gaussian yield-error scale. This truth-assisted control diagnoses absorption; it is not a usable data fit or a measured upper-limit gain.

## READ THE PLOT

Remove injected signal only from the GP training bins; keep fit counts, template and mask fixed. Mean-loss errors are paired standard errors; precision ratios use 2,000 whole-toy bootstrap resamples.

## SUGGESTED EXPLANATION

For an exact MC extraction template, contamination explains almost all the mean response deficit in these tests.

## CLAIM BOUNDARY

The Gaussian fit can still lose response with clean training because its shape/full-yield convention mismatches MC. This is a controlled diagnostic, not a new limit.

# Walk through the observed 67 MeV extraction

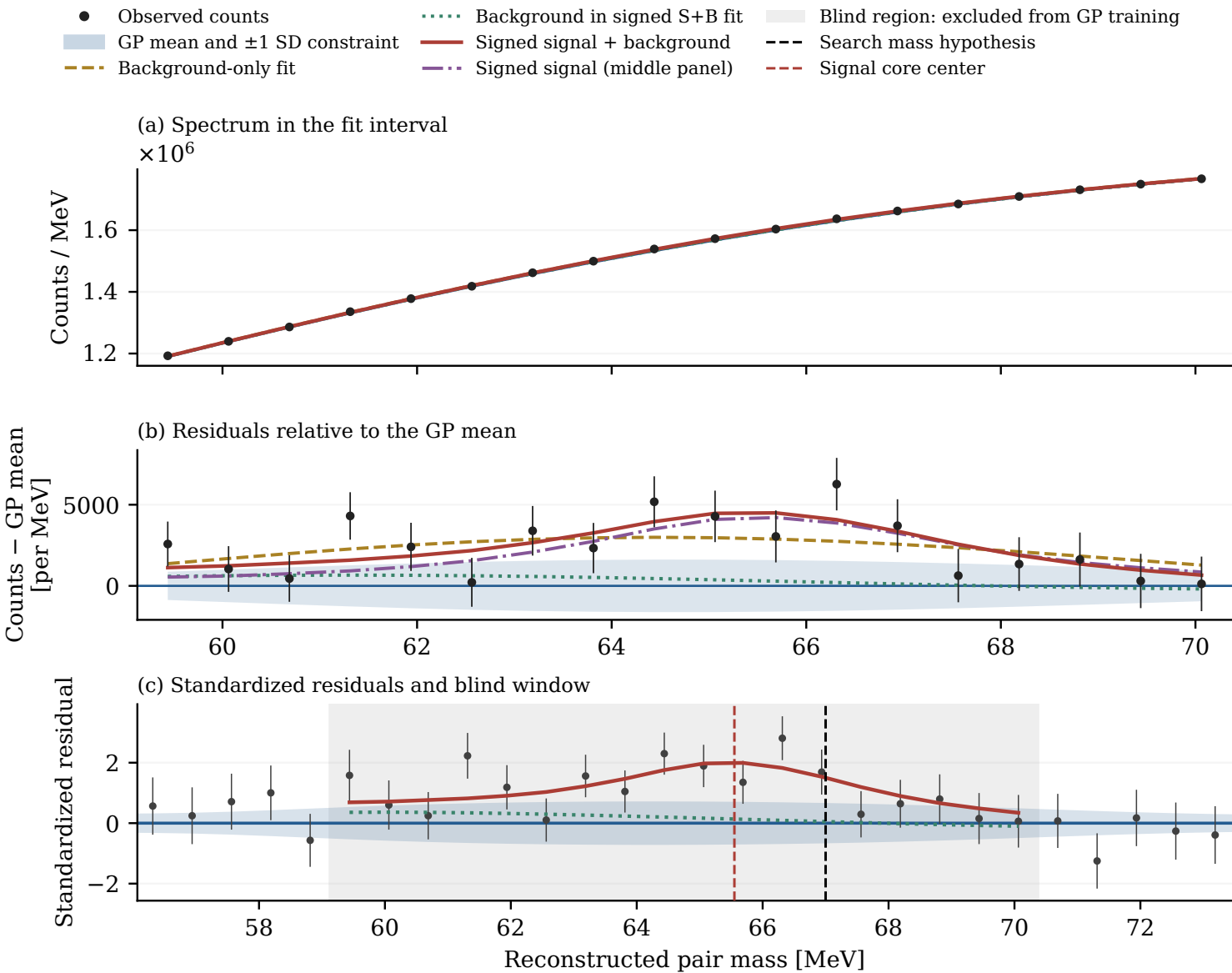
The selected MC fit has local asymptotic  $p=0.00311$ ; 1/1,000 conditional null toys exceed it.

2021 10%: selected excess 1 at a 67 MeV signal hypothesis

v16 neighboring signal-MC template |  $c = 65.551$  MeV,  $s = 1.581$  MeV

Signed full-selected yield =  $44,252 \pm 16,175$ ; local asymptotic  $Z = 2.74 \sigma$ ,  $p = 0.00311$

GP background-only toys: local excess rank  $p = 0.002$  (1 / 1000 more extreme toys); count 95% interval  $[2.53e-05, 0.00556]$



## READ THE PLOT

Read counts, data-minus-GP residuals and standardized context together. The hypothesis is 67 MeV; the template core is about 65.55 MeV. Full selected probability in training is about 47.34% here.

## SUGGESTED EXPLANATION

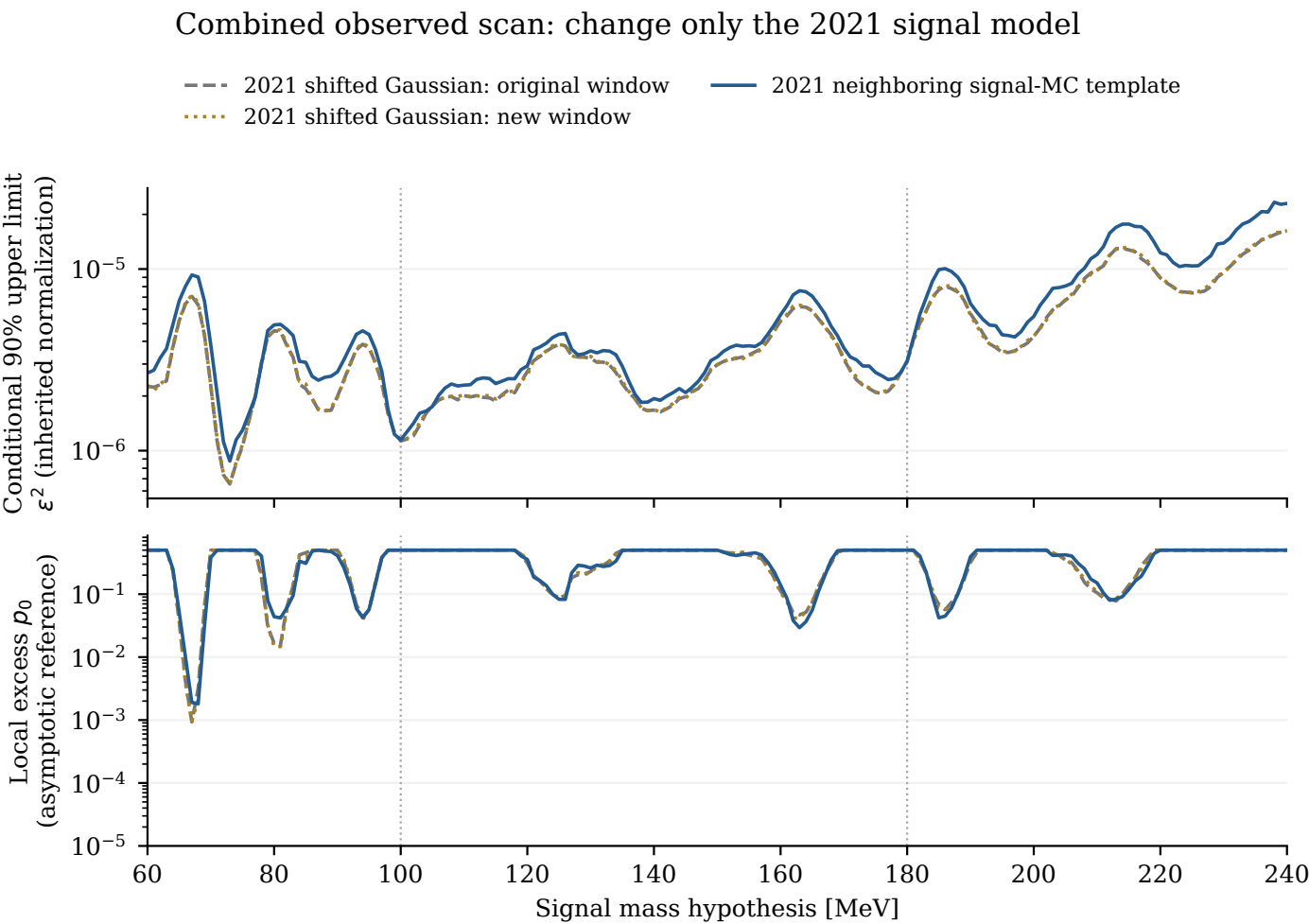
This example shows what the signal and profiled background each contribute to the same data.

## CLAIM BOUNDARY

The selected region is local and post-selection. Rank  $p=2/1001$  has exact 95% interval  $[0.0000253, 0.00556]$ ; it is not a scan-global probability.

# Follow the joint scan when only 2021 changes

The MC-method common-coupling local minimum is 68 MeV: Z=2.910, p=0.001806.



READ THE PLOT

Compare 2021 neighboring-MC/[ $-4,+3$ ]u with the shifted Gaussian in old and new windows. The 2015 and 2016 models remain the earlier ones.

SUGGESTED EXPLANATION

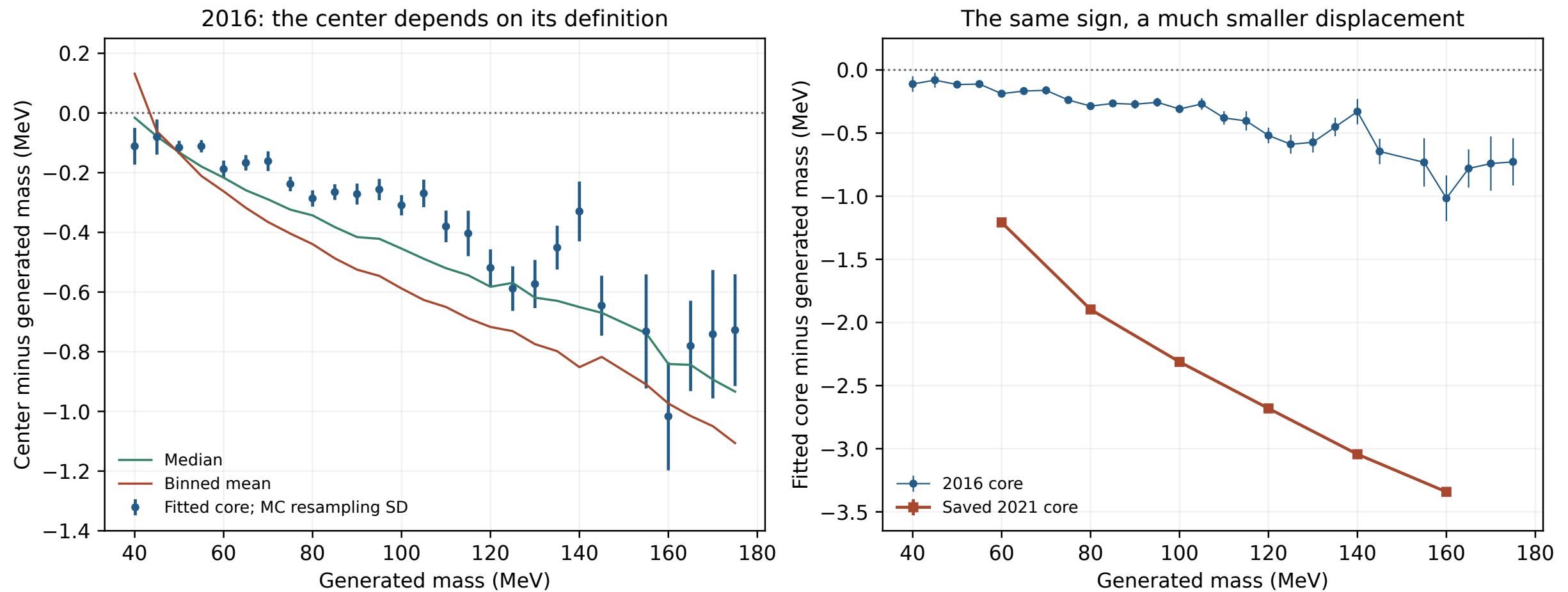
The joint fit uses one coupling with independent campaign background nuisances.

CLAIM BOUNDARY

This is an observed conditional comparison. Its 0/1,000 fixed-mass exceedances imply a finite-sample bound, not zero probability or global evidence.

# Measure the 2016 shift rather than importing 2021

Qualified 2016 cores shift down by 0.08-1.02 MeV, less than the saved 2021 shifts.



READ THE PLOT

Core, median and full mean are different location summaries. The comparison uses the supplied histogram already smeared/scaled with FEE corrections.

SUGGESTED EXPLANATION

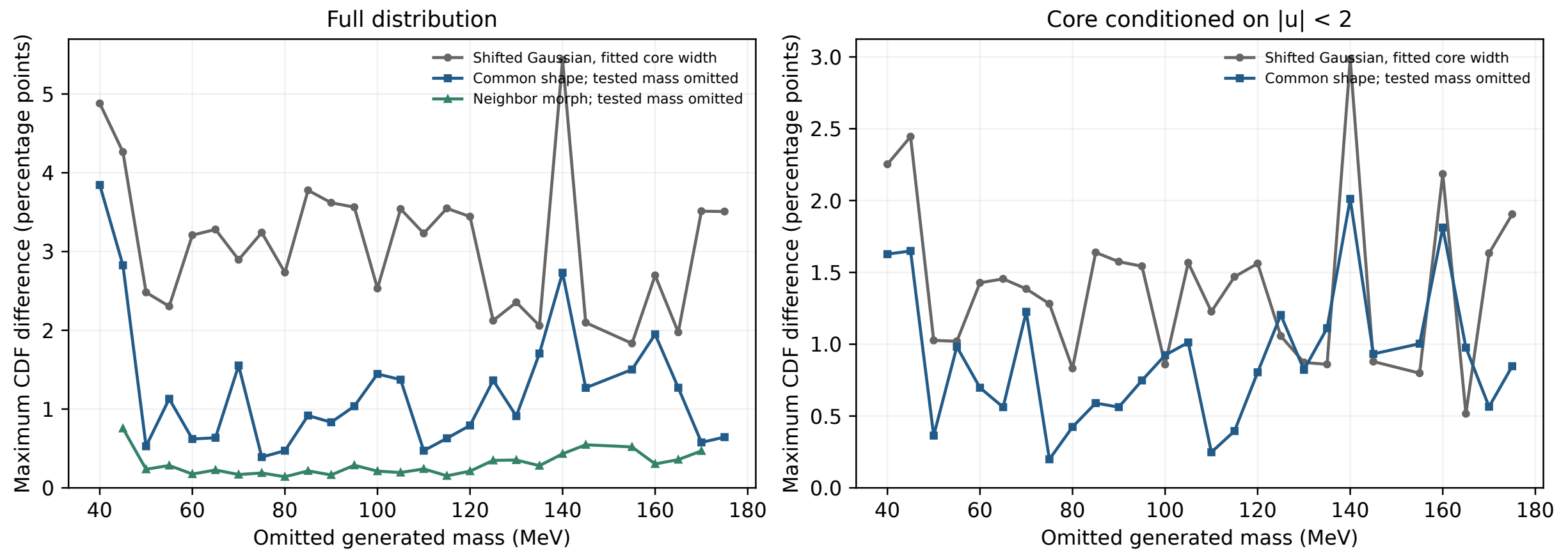
The sign is similar across campaigns, but the 2021 shift law is not a 2016 prescription.

CLAIM BOUNDARY

MC bin-resampling errors and fit-definition sensitivity are separate. The 30 and 35 MeV samples remain diagnostic-only.

# Check the interpolation on omitted masses

Neighboring-template interpolation gives held-out CDF discrepancies of 0.14-0.76 percentage points.



READ THE PLOT

Compare the shape discrepancy of an omitted native sample with neighboring-CDF interpolation, a common empirical shape, and Gaussian descriptions. Centers/widths are predicted for the held-out sample.

SUGGESTED EXPLANATION

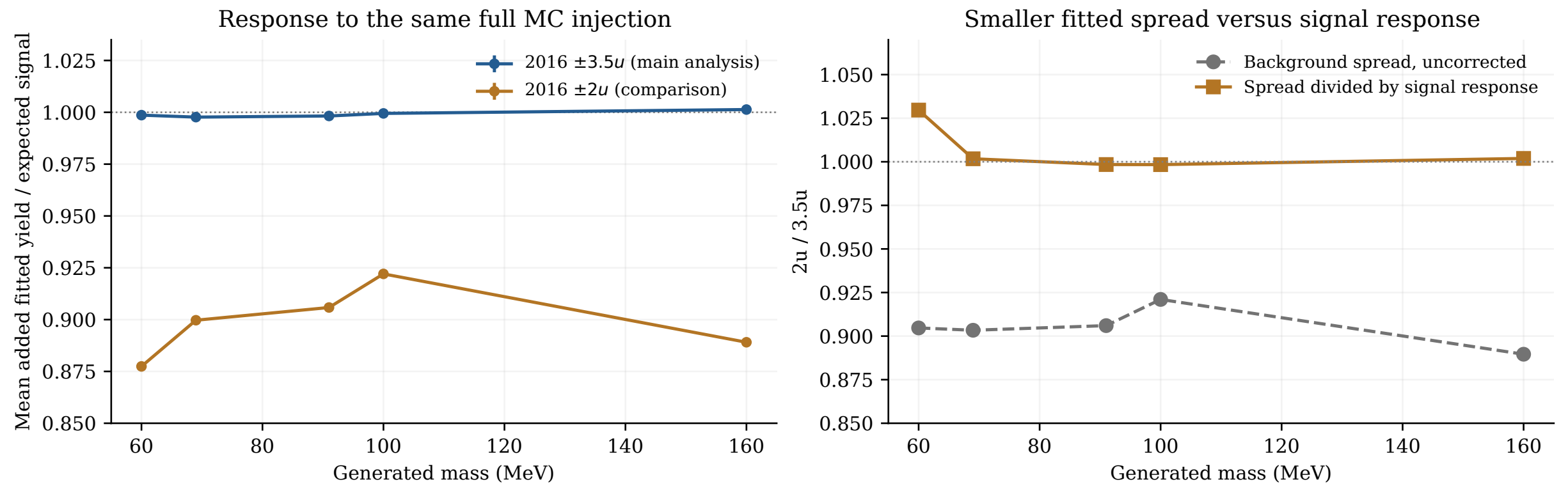
The neighboring empirical shapes are a defensible full-shape starting point within the supplied grid.

CLAIM BOUNDARY

Descriptive MC discrepancy is not a calibrated inference error or detector-systematics estimate. No 150 MeV sample was supplied.

# Test whether a smaller limit really improves sensitivity

Narrowing +/-3.5u to +/-2u lowers response; response-adjusted precision changes little.



READ THE PLOT

Five masses with 200 paired experiments give responses 0.877-0.922 versus 0.998-1.001. The spread/response ratio between windows is 0.998-1.030; response bars are standard errors of paired means.

SUGGESTED EXPLANATION

The lower raw fit spread is largely canceled by the loss of signal response.

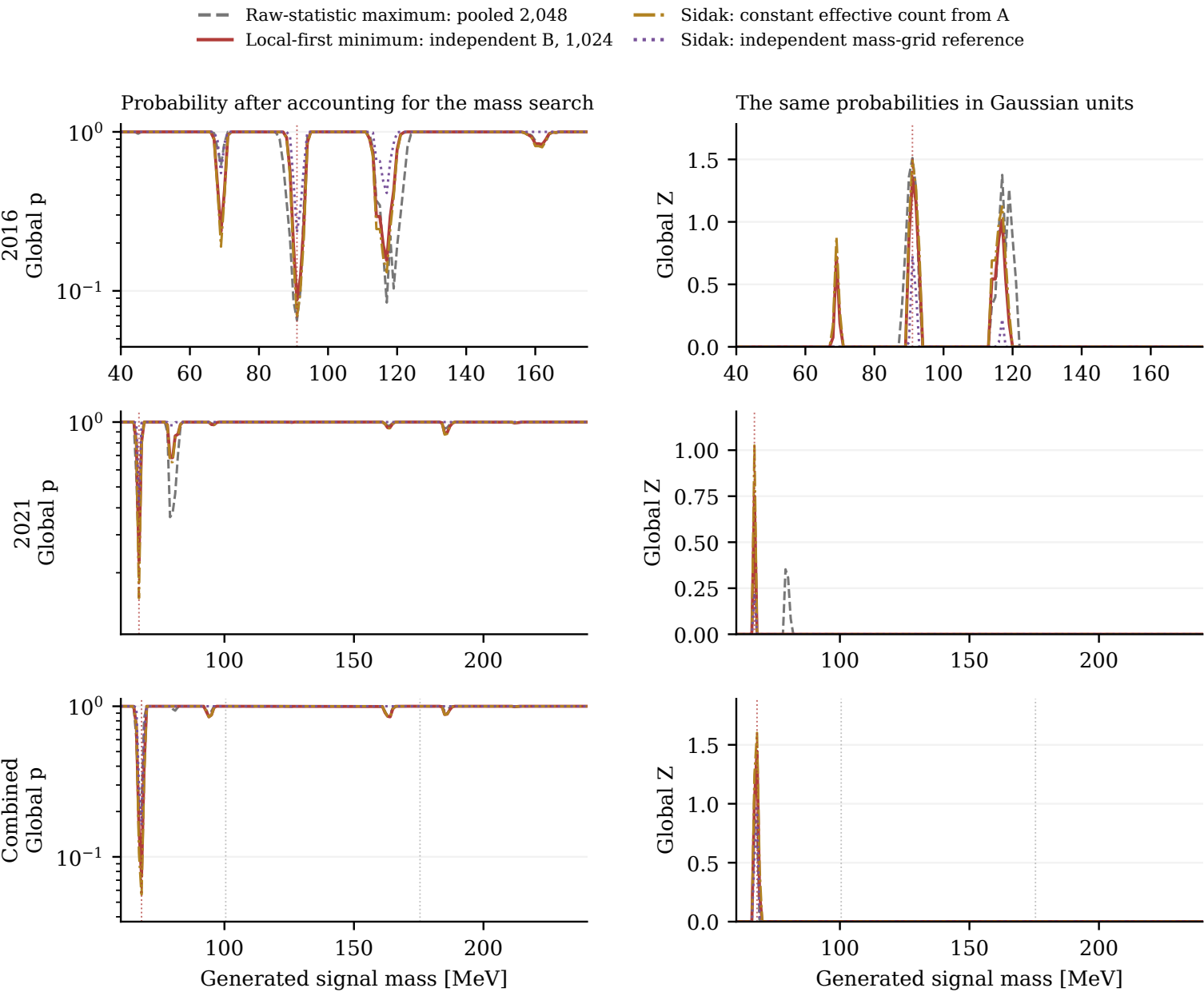
CLAIM BOUNDARY

The median observed limit falls 9.64%, but that alone is not a sensitivity gain. This is not the later +/-2.5u global protocol.

# Lead with the independently checked global probability

Combined: 75/1,024 B scans, add-one  $p=0.07415$  and 95% interval  $[0.05804,0.09095]$ .

Two search statistics, two distinct Sidak references



READ THE PLOT

A fixes mass-dependent local rank maps; independent B tests the minimum mapped  $p$  across the declared grid. 2016:  $p=0.08780$ ; 2021:  $p=0.22634$ . Intervals are exact two-sided 95% Clopper-Pearson count intervals.

SUGGESTED EXPLANATION

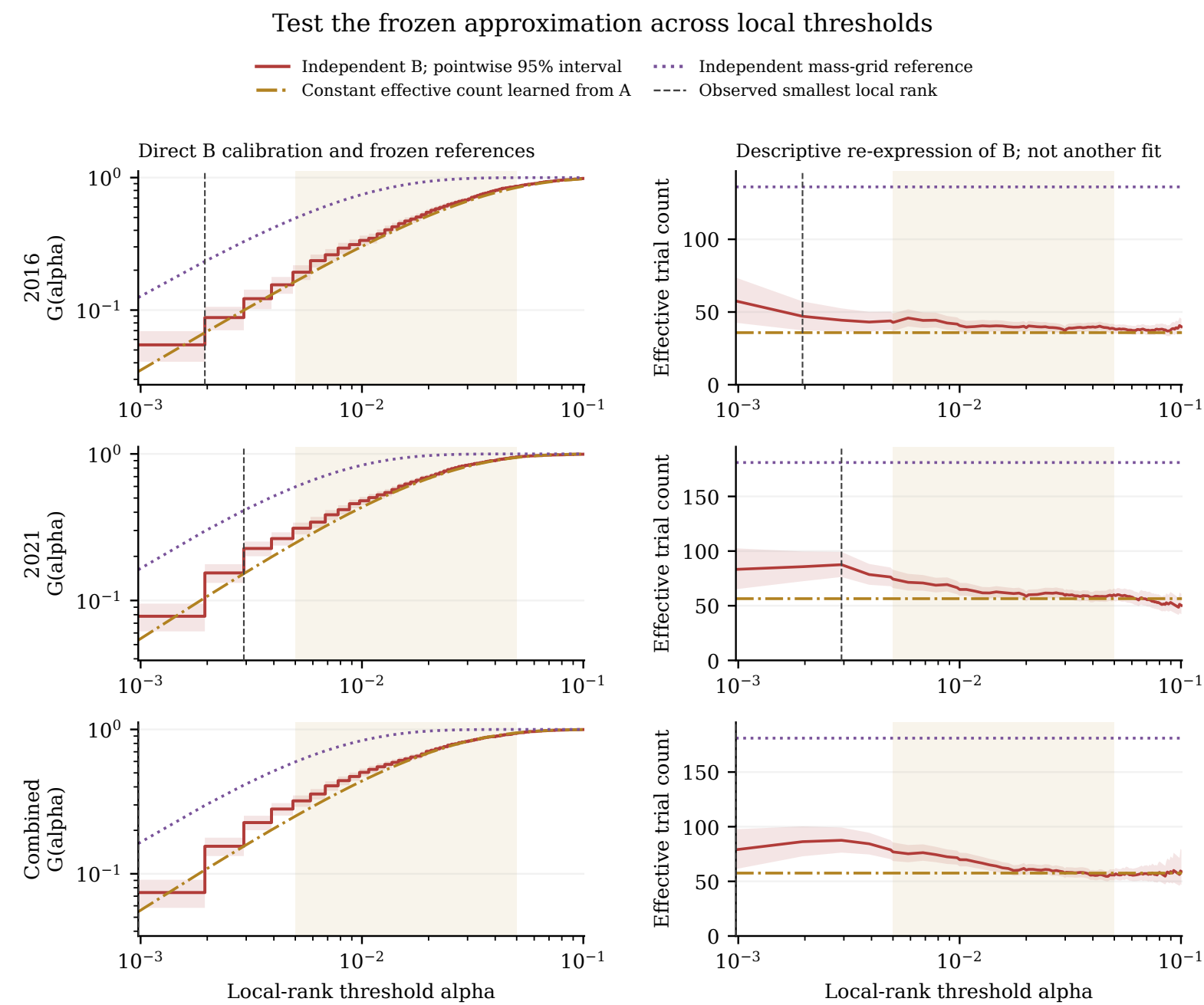
The common-coupling result is a modest conditional scan-wide excess, about 1.45 Gaussian-equivalent sigma.

CLAIM BOUNDARY

Combined is one joint coupling, not a choice among three searches. The A-map floor is  $1/1025$ ; B ties count. Source/map uncertainty and prior selection are omitted.

# Treat effective-trial fits as comparisons

At the observed minima, the fitted Sidak curves lie below the direct B intervals.



READ THE PLOT

One effective count was fitted using A-only thresholds 0.005-0.05; the observed minima lie below that range. B supplies an independent check of the resulting approximation.

SUGGESTED EXPLANATION

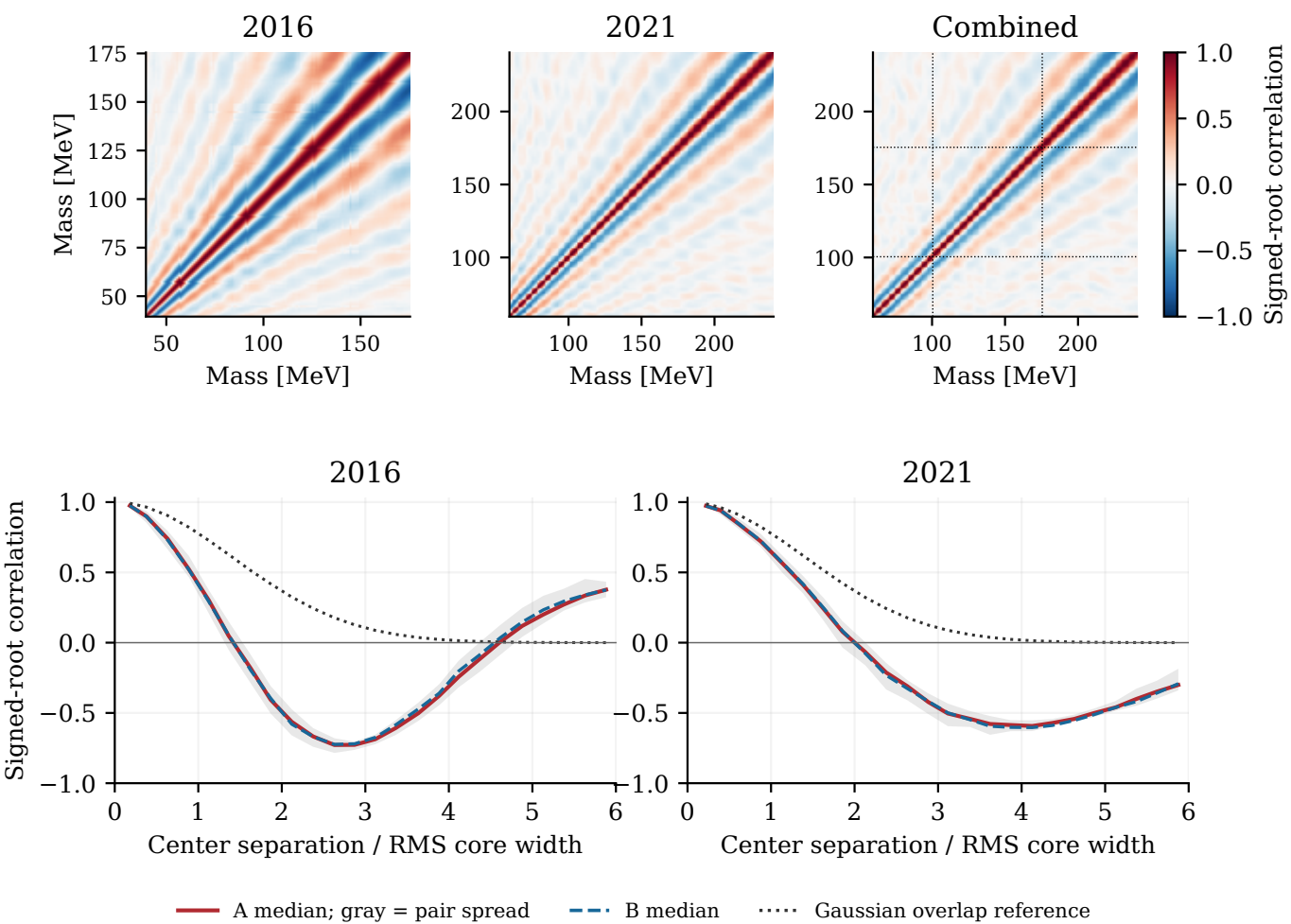
A compact trial-factor formula is useful only after its tail behavior is checked against complete scans.

CLAIM BOUNDARY

The effective count is threshold-dependent. Neither the fitted curve nor independent-grid Sidak replaces the direct B result.

# Show why neighboring tested masses are dependent

Overlapping masks, signal widths, tails and GP conditioning create correlated scan behavior.



READ THE PLOT

Read the mass-correlation matrices and separation measured in MC core-width units. The summary bands show variation across mass pairs.

SUGGESTED EXPLANATION

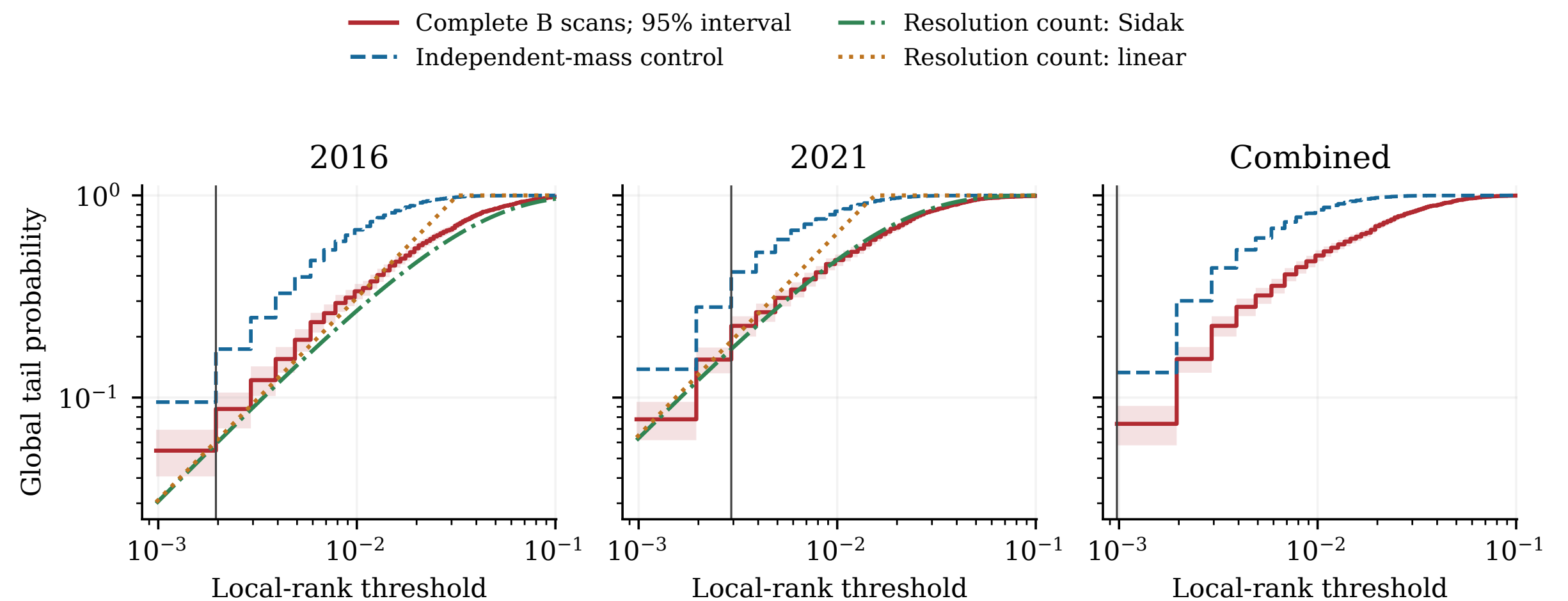
The full-scan toys carry these dependencies automatically; counting mass points discards them.

CLAIM BOUNDARY

Bands over pairs are not confidence intervals. A single resolution count cannot represent every tail threshold or the changing combined model.

# Compare the real scan with an independence control

Combined direct  $p=0.074$ , versus 0.133 when empirical mass dependence is broken.



READ THE PLOT

The independence control preserves empirical local marginals while changing their dependence. Single-year MC-resolution Sidak gives 0.059 (2016) and 0.174 (2021), below the direct 0.088 and 0.226.

SUGGESTED EXPLANATION

Correlations reduce the penalty relative to this independence control, but the simple resolution approximation reduces it too far.

CLAIM BOUNDARY

This appendix reuses the unchanged A/B arrays; it is not a second independent calibration. No new upper limit is computed.

# Context and earlier significance

Full explanations and links are in STUDY\_LOG.md; every catalogue entry has a provenance-backed package.

## World contours and exposure projections

v5p0p4\_figure2\_publication / Sep 13

## Common coupling and alternative rate models

v5p5p4\_combination\_rate\_bands / Sep 13

## Angular acceptance illustration

v5p7p0\_hps\_acceptance / Sep 13

## Local mapping, stress response and spacing

v5p8p0\_local\_significance\_mapping / Sep 17

## Coherent nominal-GP significance fields

v5p8p2\_nominal\_gp\_significance / Sep 17

## Combination methods and window widths

v5p8p4\_independent\_combinations\_windows / Sep 18

## Consolidated calibration review

v5p8p5\_consolidated\_fairness / Sep 21

## Fixed background inside the likelihood

v5p8p5\_fixed\_background\_significance / Sep 21

## Fixed-mass offset versus positive scan maximum

v5p9p5\_null\_bias / Sep 22

## Consolidated analysis note v5.0.5

v5p0p5\_analysis\_note / Sep 16

## Projected upper limits and GP echoes

v5p6p1\_2021\_upper\_limit\_echoes / Sep 13

## Geometry and magnetic transport

v5p7p1\_geometry\_acceptance / Sep 13

## Alternative background sources

v5p8p1\_background\_truths / Sep 17

## Resolution and global-probability explanation

v5p8p3\_global\_interpretation / Sep 17

## Common mass with independent amplitudes

v5p8p5\_mass\_coherence / Sep 19

## Saved-result statistical perspective

v5p8p5\_external\_statistician / Sep 21

## Unshifted local and raw-maximum global scans

v5p8p5p3\_raw\_significance / Sep 21

## APEX contour comparison

apex\_initial\_studies

# Signal templates, recovery and observed fits

Full explanations and links are in STUDY\_LOG.md; every catalogue entry has a provenance-backed package.

## Controlled Gaussian-tail perturbations

v5p9\_tail\_structure / Sep 21

## Exact-N MC injection and recovery

v6p2\_mc\_injection / Sep 23

## 100-pilot / 100-evaluation 2021 study

v6p3p1\_fixed\_yield\_2021\_100toy / Sep 24

## Unified 2021 procedure

v6p3p5\_unified\_2021 / Sep 24

## v16 templates and fit-window study

v6p3p7\_template\_windows / Sep 25

## Observed 2021 morphed-template extraction

v6p3p8\_observed\_morph / Sep 25

## 2016 already-smeared MC catalogue

v6p4\_2016\_mc\_shapes / Sep 25

## 2016 narrower-window comparison

v6p4p2\_2016\_window\_comparison / Sep 25

## Native 2021 MC shapes and windows

v6p1\_mc\_signal\_templates / Sep 22

## Fixed expected yield and reference matching

v6p3\_injection\_design / Sep 23

## 2016 exposure and calibration transfer

v6p3p2\_2016\_offset\_transfer / Sep 24

## Standalone 2021 reader's note

v6p3p6\_readability\_2021 / Sep 24

## Controlled training-contamination test

v6p3p7\_window\_leakage\_followup / Sep 25

## Joint scan with 2021 MC only

v6p3p9\_combined\_morph / Sep 25

## 2016 and joint MC extraction

v6p4p1\_shifted\_mc\_extraction / Sep 25

# Global calibration and presentation history

Full explanations and links are in STUDY\_LOG.md; every catalogue entry has a provenance-backed package.

## Complete-scan raw-statistic calibration

v6p4p3\_global\_mc / Sep 25

## Raw versus local-calibrated ordering audit

v6p4p3\_global\_method\_review / Sep 25

## Independent local-to-global calibration

v6p4p4\_calibrated\_local\_global / Sep 25

## Mass correlation and the look-elsewhere effect

v6p4p5\_mass\_correlation\_lee / Sep 26

## Initial presentation revision

unblind\_meeting\_RCmeet / Sep 22

## Presentation validation and speaker scripts

unblind\_meeting\_RCmeet / Sep 23

## Latest saved presentation follow-up

unblind\_meeting\_RCmeet / Sep 23b

## Archive and reproducibility

The publication snapshot preserves 40 study/presentation entries, complete saved inputs and ledgers, and all member payloads of 53 delivery ZIPs. Large raw files are stored losslessly compressed. The archive README explains verification, restoring large files, and rebuilding ZIP containers with identical member bytes.

This logbook is rebuilt from saved artifacts only. No new scientific fits or toys were run for it. The original study workspace remains separate from the publication checkout. Figures remain linked to original sources, with selected PNG/PDF copies and hashes for presentation reuse.